Mediant Trees for Rational Approximation and Sturmian Word Generation

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Abstract

Given an oracle for a real number θ of sufficient precision, we present four contributions. First, an algorithm that constructs a *mediant tree* of increasingly precise rational approximations to θ: starting from a Farey-adjacent bounded pair bracketing θ, each iteration applies a weighted mediant step with an optimal integer weight w^+ given by a closed-form formula (Theorem 8.1). The denominator structure mirrors the classical continued-fraction recurrence (Section 10); exponential denominator growth is proved in Theorem 10.9. Second, a Sturmian transduction that converts the mediant tree into a *mediant word* — a string over $\{\mathsf{O}, \mathsf{E}\}$ encoding the full approximation path — and establishes that this word is identical to the Sturmian word with slope $1/(1+\theta)$ at all interior positions, with the terminal symbol of each finite prefix fully characterised by whether r is an overestimate or underestimate of θ (Theorem 12.6). Third, we prove two structural invariants of mediant words: every non-leaf word begins with EE (Theorem 12.4), and the interior slice $\pi(r)[1{:}L{-}1]$ of any mediant word of length $L \geq 2$ is a palindrome (Theorem 12.5). Fourth, we show how to extend both algorithms to real numbers $\theta \in (0, 1)$ and $\theta < 0$ via simple algebraic reductions to the $\theta > 1$ case.

DRAFT

Contents

1 Introduction **3**

I Background **5**

2 The Stern–Brocot Tree and Mediants **5**
Definition 2.1 (Farey adjacency) 6
Definition 2.2 (Convergents and semi-convergents) 6

3 Farey Sequences **6**

4 Sturmian Sequences **7**
Definition 4.1 (Sturmian word) 7

5 Christoffel Words **7**
Definition 5.1 (Christoffel words) 7
Theorem 5.2 (Christoffel concatenation (Berstel [5], Theorem 1.5.1)) 7

II The Algorithm **8**

6 Definitions and Notation **8**
Definition 6.2 (Ratio) 8
Definition 6.3 (Mediant) 8
Definition 6.4 (Good estimate) 9
Definition 6.5 (Bounded pair) 9
Definition 6.6 (λ and ε) 9
Definition 6.7 (Δ and Θ) 9

7 The Mediant Construction **10**
Theorem 7.1 (Mediant convergence) 10
Corollary 7.2 (Straddling inequality) 11

8 The Optimal Weight Formula **11**
Theorem 8.1 (Optimal weight) 11
Corollary 8.2 (Children at w^+ form a bounded pair) 12

9 Mediant Words and the Mediant Tree **13**
9.1 Leaf words 14
Definition 9.1 (Leaf mediant word) 14
9.2 Branch words and the weighted operators 14
Definition 9.2 (Branch mediant word) 14
9.3 Mediant words are Christoffel words 15
Lemma 9.3 (Every mediant word is a lower Christoffel word) 15
9.4 The mediant tree 16
Definition 9.5 (Mediant tree) 16
9.5 The mediant word encodes the fraction 17

10 Convergence Analysis **17**
Theorem 10.1 (Strict decrease of $|\varepsilon|$) 17
Corollary 10.2 (Convergence to θ) 17
Theorem 10.3 (MT fractions lie on the Stern–Brocot path) 18
Theorem 10.4 (MT never skips a convergent) 18
Theorem 10.5 (Semi-convergent characterisation of r_{2d+1}) 19
Theorem 10.7 (Semi-convergent characterisation of r_{2d}) 20
Theorem 10.9 (Linear depth bound) 20
Conjecture 10.10 (Sub-universal bound) 21
Conjecture 10.11 (Worst case at ϕ) 21

11 Examples and Computational Results **23**

DRAFT

12 Relationship to Sturmian Sequences **23**
12.1 Structural properties of mediant words 23
Theorem 12.1 (Every mediant word ends in O) 23
Theorem 12.2 (Overestimate criterion) 24
Theorem 12.4 (Universal EE prefix) 24
Theorem 12.5 (Palindrome invariant) 25
Theorem 12.6 (Sturmian identity) 25
Corollary 12.7 (Mismatch characterisation) 26

13 Extension to $\theta \leq 1$ **27**
13.1 Case $0 < \theta < \frac{1}{2}$ 28
13.2 Case $\frac{1}{2} < \theta < 1$ 28
13.3 Case $\theta < 0$ 28
13.4 Degenerate cases $\theta \in \{0, \frac{1}{2}, 1\}$ 29

14 Conclusion and Open Problems **29**

A Proof of Theorem 8.1 **31**

B Examples and Computational Results **32**
B.1 Worked example: mediant words for $\log_2 3$ 32
B.2 Approximating $\log_2 3$ 33
B.3 Approximating π 34

Disclosures **34**

Reviewer's Comments **35**

DRAFT

1 Introduction

The approximation of irrational numbers by rationals is a classical subject in number theory. The Stern–Brocot tree and Farey sequences provide a systematic account of best rational approximants, and the continued-fraction expansion of a real number yields its convergents — the fractions closest to the target with bounded denominator. Sturmian words, introduced by Morse and Hedlund [1], give a combinatorial counterpart: the infinite binary sequence $s_n = \lfloor (n+1)\alpha \rfloor - \lfloor n\alpha \rfloor$ with slope $\alpha \in (0, 1)$ is the lower mechanical (Sturmian) word of slope α with intercept 0, having O-density α (other shifts are also balanced and aperiodic with the same density).

The present paper makes four contributions to this classical picture.

1. **Optimal weight formula.** We introduce the *bounded-pair* framework, in which two Farey-adjacent rational estimates bracket θ from above and below. Each weighted mediant step is a run of w steps in the Stern–Brocot tree in the same direction — the classical construction (see Graham, Knuth and Patashnik [2]). What is new is Theorem 8.1: a closed-form expression for the unique *optimal* integer weight w^+ —

the one whose mediants straddle θ — derived from a sign-change bracket on the linear error function, enabling efficient iteration with no redundant steps.

2. **Mediant word representation.** The mediant tree can be encoded as a string over $\{\mathsf{O}, \mathsf{E}\}$, the *mediant word* (Section 9). This representation connects rational approximation to combinatorics on words.

3. **Sturmian identity.** The mediant word produced by the mediant tree is identical to the Sturmian word with slope $1/(1+\theta)$ (Theorem 12.6), agreeing at every position except possibly the last symbol of each finite prefix. The identification of Christoffel words with Sturmian prefixes is classical (Lothaire [3]); the novelty is the direct proof via the mediant construction and the precise last-letter criterion.

4. **Compressed transducer.** Each iterate produced by the algorithm is a convergent or semi-convergent of θ (a node on the Stern–Brocot path of θ) — the best bracketing fraction achievable in one step from the current bounded pair. The mediant tree is therefore a *compressed transducer* for the Sturmian word: the approximation path ($O(\log N)$ real-arithmetic iterations, each requiring $O(1)$ oracle evaluations; proved via exponential denominator growth, Theorem 10.9) encodes the entire construction history, after which the N symbols are enumerated by a purely integer recursive traversal with no further real arithmetic.

The algorithm requires an oracle for θ — a means of evaluating $\lceil o\theta \rceil$ to sufficient precision at each step — but is otherwise entirely general. Given such an oracle, the mediant tree can represent *any* real number as a mediant word over $\{\mathsf{O}, \mathsf{E}\}$, together with a tag identifying the regime of Table 1, making it a universal encoder from reals to binary words.

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Organisation. Part I (Sections 2–5) collects background on the Stern–Brocot tree, Farey sequences, Sturmian words, and Christoffel words. Part II develops the algorithm and its theory: Section 6 introduces the core definitions; Section 7 develops the mediant construction and straddling inequality; Section 8 derives the optimal weight formula; Section 9 introduces mediant words; Section 9.4 describes the mediant tree; Section 10 analyses the rate of convergence; Section 11 presents examples and computational results; Section 12 establishes the connection to Sturmian words; Section 13 extends the construction to all real θ; Section 14 concludes with open questions. Appendix A contains the proof of Theorem 8.1.

Notation. Throughout, $\theta > 1$ is the target real number (the case $\theta \notin (1, \infty)$ is handled by reduction in Section 13). A rational approximant is written $r = e/o$ with numerator e (*even-count*) and denominator o (*odd-count*); $e, o > 0$ always, and $\gcd(e, o) = 1$.

Alphabet. Mediant words are strings over $\{\mathsf{E}, \mathsf{O}\}$, where E stands for *even* and O for *odd*.

Depth index. The algorithm progresses through depths $d = 0, 1, 2, \ldots$. At each depth d the algorithm holds a *bounded pair* (r_{2d}, r_{2d+1}), where r_{2d+1} is always the better estimate (smaller relative error $|\varepsilon|$) and r_{2d} the worse. The flat linear subscripts $2d$ and $2d+1$ reflect the sequential history of all estimates produced: r_0, r_1 are the initial integer bounds, and

each depth $d \geq 1$ contributes two new estimates r_{2d} and r_{2d+1} (one of which may coincide with r_{2d-2} when $w^- = 0$).

Intermediate quantities (depth d to $d+1$). Between depth d and depth $d+1$ the algorithm computes several quantities that are intermediate in the sense that they are derived from the pair (r_{2d}, r_{2d+1}) and used to produce the next pair. These carry *superscript* decorations $+$, $-$ (or $*$ for a combined value) to distinguish them from the depth-indexed estimates:

- w^+: the *optimal weight* — the unique integer such that $\text{Mediant}(r_{2d}, r_{2d+1}, w^+ - 1)$ and $\text{Mediant}(r_{2d}, r_{2d+1}, w^+)$ straddle θ.
- $w^- = w^+ - 1$: the companion (sub-optimal) weight.
- r_+, r_-: the two candidate mediants at weights w^+ and w^-; the better one becomes r_{2d+3} and the worse r_{2d+2}.
- ε^+, ε^-: the relative errors of r_+ and r_-.
- λ_+, λ_-: the ratios $(e/o)/\theta$ for r_+ and r_-.
- e_+, e_- and o_+, o_-: the numerators and denominators of r_+ and r_-.

The superscript/subscript distinction is systematic: a $+/-$ *superscript* always marks an intermediate quantity between two depths; a numeric *subscript* always marks a position in the flat estimate sequence. These two systems are complementary and never overlap. When the depth matters we write $w_d^\pm$ for the propagated weights of Definition 9.2; in §§7–8 the depth is fixed and the subscript is omitted.

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Per-estimate scalar quantities. For a single estimate $r = e/o$ the key scalars are:

- $\lambda(r) = (e/o)/\theta$: the ratio of r to θ; equals 1 iff $r = \theta$.
- $\varepsilon(r) = \lambda(r) - 1$: the *relative error*; negative for underestimates, positive for overestimates.
- $\Delta(r, r') = e'o - eo'$: the *Farey determinant* of two estimates; $|\Delta| = 1$ iff r and r' are Farey adjacent.
- $\Theta(r) = o(e - \theta o)$: the *scaled error* of estimate $r = e/o$ (Definition 6.7); for the current pair, $\Theta := \Theta(r_{2d+1})$. Equals $\theta\, o^2\, \varepsilon$ and $\Theta = 0$ iff $r = \theta$ exactly.

Part I
Background

2 The Stern–Brocot Tree and Mediants

The *Stern–Brocot tree* is a complete binary search tree containing every positive rational exactly once in lowest terms (see, e.g., Graham, Knuth, and Patashnik [2]). It is constructed

by repeated *mediant* insertion: starting from the pair $(0/1, 1/0)$, the mediant of two adjacent fractions a/b and c/d is $(a+c)/(b+d)$, and inserting it between them yields two new adjacent pairs. The Stern–Brocot tree encodes continued-fraction expansions as paths from the root: a partial quotient of k corresponds to k steps in the same direction before the first reversal.

Definition 2.1 (Farey adjacency)**.** Two fractions a/b and c/d (in lowest terms, $a/b < c/d$) are *Farey adjacent* if

$$bc - ad = 1.$$

Equivalently, they are adjacent in some Farey sequence F_n, or they lie on adjacent branches of the Stern–Brocot tree. Farey adjacency is equivalent to the condition that no rational with denominator at most $\max(b, d)$ lies strictly between a/b and c/d.

The mediant $(a + c)/(b + d)$ of two Farey-adjacent fractions is itself Farey adjacent to each of them. This adjacency-preservation is the key structural property exploited by our algorithm.

The path from the root of the Stern–Brocot tree to θ passes through the *convergents* p_k/q_k of θ, defined as the fractions produced by truncating the continued-fraction expansion $\theta = [a_0; a_1, a_2, \ldots]$. The nodes visited between consecutive convergents p_{k-1}/q_{k-1} and p_k/q_k are the *semi-convergents*:

Definition 2.2 (Convergents and semi-convergents)**.** Let $\theta = [a_0; a_1, a_2, \ldots]$ with $p_{-1}/q_{-1} = 1/0$ and convergents p_k/q_k for $k \geq 0$. The *semi-convergents* between consecutive convergents p_k/q_k and p_{k+1}/q_{k+1} are the fractions

$$\frac{p_{k-1} + j\, p_k}{q_{k-1} + j\, q_k}, \quad 1 \leq j < a_{k+1}.$$

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(These appear on the SB path strictly between p_k/q_k and p_{k+1}/q_{k+1}.) The integer spine $1/1, 2/1, \ldots, (\lfloor\theta\rfloor - 1)/1$ consists of the initial SB-path nodes strictly before the first convergent $p_0/q_0 = \lfloor\theta\rfloor/1$. The convergents and semi-convergents together, plus the integer spine, form exactly the nodes on the Stern–Brocot path of θ. We use this broad definition throughout; in particular every SB-path node that is not a convergent is called a semi-convergent, with no additional size condition on j.

3 Farey Sequences

The *Farey sequence* F_n is the increasing sequence of fractions in $[0, 1]$ with denominator at most n, arranged in order. Its fundamental property is that consecutive terms $a/b < c/d$ in F_n satisfy $bc - ad = 1$ (i.e. they are Farey adjacent), and their mediant $(a + c)/(b + d)$ is the fraction with smallest denominator strictly between them.

For our purposes the key fact is the following: if a/b and c/d are Farey adjacent and $\theta \in (a/b, c/d)$, then every rational strictly between a/b and c/d with denominator at most $b+d$ equals their mediant $(a+c)/(b+d)$. This is why maintaining Farey adjacency throughout the algorithm ensures that no improvement step is redundant.

4 Sturmian Sequences

Sturmian words were introduced by Morse and Hedlund [1] as the simplest aperiodic symbolic sequences. Let $\alpha \in (0,1)$ be irrational. The *lower Sturmian word* with slope α is defined by

$$s_n = \lfloor (n+1)\alpha \rfloor - \lfloor n\alpha \rfloor, \qquad n \geq 0.$$

Encoding $s_n = 1$ as O and $s_n = 0$ as E, this is a binary sequence with O-density exactly α.

Definition 4.1 (Sturmian word)**.** The *Sturmian word* $\mathbf{s}(\alpha)$ with slope $\alpha \in (0,1)$ is the infinite word over $\{\mathsf{O}, \mathsf{E}\}$ defined by $s_n = \mathsf{O}$ if $\lfloor (n+1)\alpha \rfloor > \lfloor n\alpha \rfloor$, and $s_n = \mathsf{E}$ otherwise.

Sturmian words are *balanced*: no two factors of the same length differ in their O-count by more than one. For irrational α, the lower mechanical (Sturmian) word of slope α with intercept 0 is the canonical balanced aperiodic sequence; shifts of it are also balanced and aperiodic [1].

The two-word structure is fundamental: a Sturmian word with slope $\alpha \in (0, 1/2)$ consists of exactly two distinct *primitive words* (unbreakable building blocks), of lengths $\lfloor 1/\alpha \rfloor$ and $\lceil 1/\alpha \rceil$, concatenated in the Sturmian pattern. Concretely, with $\alpha = 1/(1+\theta)$ for $\theta > 1$, the two primitive words are $\mathsf{E}^{\lfloor \theta \rfloor}\mathsf{O}$ and $\mathsf{E}^{\lceil \theta \rceil}\mathsf{O}$ — exactly the leaf words of the mediant tree (Section 9.1).

Since $\alpha < 1/2$ for $\theta > 1$, the sequence avoids OO: two consecutive O's would require $\lfloor (n+2)\alpha \rfloor - \lfloor n\alpha \rfloor = 2$, but $2\alpha < 1$ makes this impossible. Conversely, when $\alpha > 1/2$ (i.e. $\theta < 1$), consecutive O's are unavoidable.

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Sturmian morphisms. The two *Sturmian morphisms* $\sigma_E \colon \mathsf{E} \mapsto \mathsf{E},\ \mathsf{O} \mapsto \mathsf{EO}$ and $\sigma_O \colon \mathsf{E} \mapsto \mathsf{OE},\ \mathsf{O} \mapsto \mathsf{O}$ map Sturmian words to Sturmian words (of a different slope in general).

5 Christoffel Words

Definition 5.1 (Christoffel words)**.** For coprime positive integers e and o, the *lower Christoffel word* $\mathrm{Ch}^-(e,o)$ is the unique primitive word of length $L = e + o$ over $\{\mathsf{E}, \mathsf{O}\}$ containing exactly e copies of E and o copies of O that ends in O, with O-positions given by

$$\mathrm{Ch}^-(e,o)[i] = \mathsf{O} \iff \lfloor (i+1)o/L \rfloor > \lfloor i \cdot o/L \rfloor.$$

The *upper Christoffel word* $\mathrm{Ch}^+(e,o)$ is the reverse of $\mathrm{Ch}^-(e,o)$: it ends in E instead of O. Both words share the same palindromic interior U: $\mathrm{Ch}^-(e,o) = \mathsf{E} \cdot U \cdot \mathsf{O}$ and $\mathrm{Ch}^+(e,o) = \mathsf{O} \cdot U \cdot \mathsf{E}$ (Berstel [5], Chapter 1).

$\mathrm{Ch}^-(e,o)$ is the lower Sturmian word of slope o/L truncated to length L and normalised to end in O; $\mathrm{Ch}^+(e,o)$ is the upper Sturmian word of the same slope, ending in E (Berstel [5], Chapter 1).

Each Christoffel word is characterised purely by its fraction o/L: given $\gcd(e,o) = 1$, the pair (e,o) is uniquely determined by $o/(e+o)$.

Theorem 5.2 (Christoffel concatenation (Berstel [5], Theorem 1.5.1)). *Let* $\mathrm{Ch}^-(e_0, o_0)$ *and* $\mathrm{Ch}^-(e_1, o_1)$ *be two lower Christoffel words with* $o_0/L_0 < o_1/L_1$ *(where* $L_k = e_k + o_k$*) satisfying the Farey adjacency condition* $o_1 L_0 - o_0 L_1 = 1$*. Then for any integer* $w \geq 1$*:*

$$\begin{aligned} \mathrm{Ch}^-(e_0, o_0)^w \cdot \mathrm{Ch}^-(e_1, o_1) &= \mathrm{Ch}^-(w\, e_0 + e_1,\ w\, o_0 + o_1), \\ \mathrm{Ch}^-(e_0, o_0) \cdot \mathrm{Ch}^-(e_1, o_1)^w &= \mathrm{Ch}^-(e_0 + w\, e_1,\ o_0 + w\, o_1). \end{aligned}$$

In the first identity the smaller-slope word is repeated; in the second the larger-slope word is repeated.

Part II
The Algorithm

6 Definitions and Notation

Throughout Sections 6–12, let $\theta > 1$ be a fixed real number. Section 13 handles all other real numbers by reduction to this case. We write $\mathbb{N}$ for the positive integers and $\lfloor\cdot\rfloor$, $\lceil\cdot\rceil$ for floor and ceiling.

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Remark 6.1 (Rational vs. irrational θ). Most of the combinatorial claims in these sections — the Christoffel word structure (Lemma 9.3), the palindrome invariant (Theorem 12.5), and the EE-prefix theorem (Theorem 12.4) — hold for all real $\theta > 1$, rational or not. The Sturmian identity (Theorem 12.6) and the last-letter theorem (Theorem 12.2) require θ to be irrational, since the Sturmian word is only well-defined for irrational slope $\alpha = 1/(1+\theta)$. The uniqueness of w^+ (Theorem 8.1) also requires irrationality. The formula $w^+ = \lfloor -A/B \rfloor + 1$ rests on $-A/B$ being non-integer: the strict bracketing condition $(A+(w-1)B)(A+wB) < 0$ holds for exactly one integer w iff $-A/B$ is not an integer. For rational θ, $-A/B$ can be an integer w^-, in which case no integer satisfies the strict condition: weights w^- and $w^+ = w^- + 1$ give product zero, and all others give a positive product. The mediant at weight w^- equals θ exactly. Step 4 of the algorithm then computes $w^+ = w^- + 1$, producing two children; by the step-5 ordering, the better estimate (which is θ exactly) becomes r_{2d+3} (the *second* member of the new pair); $\Theta = 0$ is detected at the next pass through step 3, at which point the algorithm terminates. For rational θ the algorithm terminates in finitely many steps (at the step where $\Theta = 0$, i.e. where r_{2d+1} is an exact representation of θ); all claims except Theorem 12.2, Theorem 12.6, Corollary 12.7, and uniqueness in Theorem 8.1 hold for every iterate up to termination.

Definition 6.2 (Ratio). A *ratio* is a 2-tuple $r = (e, o) \in \mathbb{N}^2$ with $\gcd(e, o) = 1$, representing the rational number e/o. A ratio $r = (e, o)$ is a *rational estimate* of θ when used in that context.

Definition 6.3 (Mediant). Given two ratios $r = (e, o)$ and $r' = (e', o')$ and a non-negative integer $w \geq 0$, the *mediant* of r and r' with weight w is the ratio

$$\mathrm{Mediant}(r,\, r',\, w) = (e + w\, e',\ o + w\, o').$$

The degenerate case $w = 0$ gives $\mathrm{Mediant}(r, r', 0) = r$. The Farey determinant of the pair (r, r') is

$$\Delta(r, r') = e'o - eo'.$$

Definition 6.4 (Good estimate)**.** An estimate e/o is a *good estimate* of θ if

$$e \in \{\lfloor o \cdot \theta \rfloor, \lceil o \cdot \theta \rceil\},$$

i.e. e/o is one of the two nearest rationals to θ with denominator o. A good estimate with $e = \lceil o\theta \rceil$ satisfies $e/o \geq \theta$ and is an *upper bound*; one with $e = \lfloor o\theta \rfloor$ satisfies $e/o \leq \theta$ and is a *lower bound*.

Definition 6.5 (Bounded pair)**.** Two good estimates e_l/o_l and e_u/o_u form a *bounded pair* if e_l/o_l is a lower bound, e_u/o_u is an upper bound,

$$\frac{e_l}{o_l} < \theta \leq \frac{e_u}{o_u},$$

and they satisfy the *Farey adjacency condition*

$$e_u \cdot o_l - e_l \cdot o_u = 1.$$

The Farey adjacency condition ensures that the pair lies on adjacent branches of the Stern–Brocot tree (Section 2) and that no rational with smaller denominator lies strictly between e_l/o_l and e_u/o_u.

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Definition 6.6 (λ and ε)**.** For an estimate e/o of θ, define the *relative ratio*

$$\lambda = \frac{e/o}{\theta}$$

and the *relative error*

$$\varepsilon = \lambda - 1 = \frac{e/o}{\theta} - 1.$$

An upper-bound good estimate satisfies $\lambda \geq 1$ (so $\varepsilon \geq 0$); a lower-bound good estimate satisfies $\lambda \leq 1$ (so $\varepsilon \leq 0$). In both cases $|\varepsilon|$ is small when o is large.

Estimates are indexed so that r_{2d} and r_{2d+1} form the bounded pair at depth d, with r_{2d+1} always the better estimate (smaller $|\varepsilon|$). Their children are r_{2d+2} and r_{2d+3}, the bounded pair at depth $d + 1$. We write λ_{2d}, ε_{2d}, etc. for the corresponding values. In a bounded pair we always orient so that $|\varepsilon_{2d}| > |\varepsilon_{2d+1}|$, i.e. r_{2d+1} is the better estimate of the two.

Definition 6.7 (Δ and Θ)**.** Given a bounded pair $(r_{2d}, r_{2d+1}) = (e_{2d}/o_{2d},\ e_{2d+1}/o_{2d+1})$ with $|\varepsilon_{2d}| > |\varepsilon_{2d+1}|$, define

$$\Delta = e_{2d+1}o_{2d} - e_{2d}o_{2d+1}$$

and

$$\Theta = o_{2d+1}(e_{2d+1} - \theta \cdot o_{2d+1}).$$

More generally, define $\Theta(r) = o(e - \theta o)$ for any estimate $r = e/o$; then $\Theta := \Theta(r_{2d+1})$ for the current pair. Geometrically, $\Theta(r)/o = e - \theta o$ is the signed vertical distance from the lattice point (o, e) to the line $e = \theta o$, and $|\Theta(r)|$ is twice the area of the triangle with vertices $(0, 0)$, $(o, \theta o)$, (o, e). Equivalently, $\Theta = \theta\, o_{2d+1}^2 \varepsilon_{2d+1}$ is the signed absolute error of r_{2d+1} scaled by its denominator squared.

The Farey adjacency condition in Definition 6.5 requires $e_u \cdot o_l - e_l \cdot o_u = 1$, which is precisely $|\Delta| = 1$ (the sign depends on which of r_{2d}, r_{2d+1} is the upper bound). One checks that $|\Theta|/o_{2d+1} = |e_{2d+1} - \theta o_{2d+1}| < 1$ (the good-estimate condition), and that $\operatorname{sgn}(\Theta) = \operatorname{sgn}(\Delta)$ throughout the iteration.

Domain restriction. The definitions in this section assume $\theta > 1$. Section 13 shows how to reduce all other real numbers to this case.

Bounded pairs and the mediant tree. A bounded pair is a purely analytic object: any two good estimates that bracket θ and satisfy the Farey adjacency condition qualify, regardless of how they were found. The mediant construction (Section 7) provides a canonical way to *generate* bounded pairs, but the concept is wider. The invariants Δ and Θ belong to the pair itself; the Christoffel word structure (Section 5) is a property of how the pair was reached via the mediant tree. In particular, $\Delta = \pm 1$ and $\operatorname{sgn}(\Theta) = \operatorname{sgn}(\Delta)$ hold for any bounded pair.

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7 The Mediant Construction

The classical *mediant* of two fractions e_{2d}/o_{2d} and e_{2d+1}/o_{2d+1} is $(e_{2d} + e_{2d+1})/(o_{2d} + o_{2d+1})$. We generalise this to a weighted family parametrised by a non-negative integer w.

Theorem 7.1 (Mediant convergence)**.** *Let (r_{2d}, r_{2d+1}) be a bounded pair for θ with $|\varepsilon_{2d}| > |\varepsilon_{2d+1}|$, and let $w \geq 0$ be a non-negative integer such that*

$$\big(\text{Mediant}(r_{2d}, r_{2d+1}, w-1),\ \text{Mediant}(r_{2d}, r_{2d+1}, w)\big)$$

brackets θ (interpreting $\text{Mediant}(r_{2d}, r_{2d+1}, -1) = r_{2d}$ when $w = 0$, so that $r^- = r_{2d}$ is the parent itself). Write $r_- = \text{Mediant}(r_{2d}, r_{2d+1}, w-1)$ and $r_+ = \text{Mediant}(r_{2d}, r_{2d+1}, w)$ for these two candidates. Then

$$|\varepsilon^-| \leq |\varepsilon_{2d}| \qquad \text{and} \qquad |\varepsilon^+| < |\varepsilon_{2d+1}| < |\varepsilon_{2d}|.$$

(When $w = 0$, $r_- = r_{2d}$ and $|\varepsilon^-| = |\varepsilon_{2d}|$; the bounds hold trivially. When $w = 1$, $r_- = r_{2d}$ exactly so $|\varepsilon^-| = |\varepsilon_{2d}|$; otherwise $|\varepsilon^-| < |\varepsilon_{2d}|$.) Moreover $|e_+ o_- - e_- o_+| = 1$, so (r_-, r_+) is Farey-adjacent. The two mediants are assigned as r_{2d+2} and r_{2d+3}, where r_{2d+3} is whichever has the smaller $|\varepsilon|$ (the better estimate), and r_{2d+2} takes the other. Since r_+ always satisfies $|\varepsilon^+| < |\varepsilon_{2d+1}|$, we have $|\varepsilon_{2d+3}| < |\varepsilon_{2d+1}|$ regardless of which candidate is assigned to r_{2d+3}.

Proof. The denominators satisfy $o_- = o_{2d} + (w-1)o_{2d+1} < o_{2d} + w\, o_{2d+1} = o_+$, so r_- and r_+ are distinct. The ratio λ_- is a weighted combination of λ_{2d} and λ_{2d+1}:

$$\lambda_- = \frac{o_{2d}\,\lambda_{2d} + (w-1)o_{2d+1}\,\lambda_{2d+1}}{o_{2d} + (w-1)o_{2d+1}}.$$

When $w = 1$ this gives $\lambda_- = \lambda_{2d}$ exactly, so $|\varepsilon^-| = |\varepsilon_{2d}|$. When $w \geq 2$, r_- lies strictly between r_{2d} and r_{2d+1}. Write $f(t) = A + tB$ where $A = e_{2d} - \theta o_{2d}$ and $B = e_{2d+1} - \theta o_{2d+1}$ are the scaled errors of r_{2d} and r_{2d+1}; then $f(w) = e_+ - \theta\, o_+$. From the bracketing hypothesis $f(w-1)f(w) < 0$ it follows that $B \neq 0$ (otherwise f would be constant and could not change sign), and that the zero $c = -A/B$ of f lies in $(w-1, w)$. Thus $f(w-1)$ has the sign of A (the side of r_{2d}) and $f(w)$ has the sign of B (the side of r_{2d+1}). Since the bracketing condition places r_- on the same side of θ as r_{2d}, the ratio r_-/θ lies between r_{2d}/θ and 1, so $|\lambda_- - 1| < |\lambda_{2d} - 1|$, i.e. $|\varepsilon^-| < |\varepsilon_{2d}|$. For r_+ (weight $w \geq 1$), the bracketing condition places r_+ on the same side of θ as r_{2d+1}, so r_+/θ lies between r_{2d+1}/θ and 1, giving $|\lambda_+ - 1| < |\lambda_{2d+1} - 1|$, i.e. $|\varepsilon^+| < |\varepsilon_{2d+1}| < |\varepsilon_{2d}|$. The Farey adjacency condition for (r_-, r_+) follows by direct expansion:

$$\begin{aligned} e_+\, o_- - e_-\, o_+ &= (e_{2d} + w\, e_{2d+1})(o_{2d} + (w-1)o_{2d+1}) - (e_{2d} + (w-1)e_{2d+1})(o_{2d} + w\, o_{2d+1}) \\ &= \big[(w-1) - w\big] e_{2d}\, o_{2d+1} + \big[w - (w-1)\big] e_{2d+1}\, o_{2d} \\ &= -e_{2d}\, o_{2d+1} + e_{2d+1}\, o_{2d} = \Delta, \end{aligned}$$

so $|e_+\, o_- - e_-\, o_+| = |\Delta| = 1$, and the Farey adjacency of (r_-, r_+) is inherited directly from that of (r_{2d}, r_{2d+1}). □

Corollary 7.2 (Straddling inequality)**.** *Under the conditions of Theorem 7.1,*

$$\varepsilon^-\, \varepsilon^+ < 0 \qquad \textit{and} \qquad |\varepsilon^-||\varepsilon^+| < |\varepsilon_{2d}|\, |\varepsilon_{2d+1}|.$$

Proof. The first inequality holds because r_- and r_+ straddle θ, so their errors have opposite signs. For the second: $|\varepsilon^-| \leq |\varepsilon_{2d}|$ and $|\varepsilon^+| < |\varepsilon_{2d+1}|$ (Theorem 7.1), so $|\varepsilon^-||\varepsilon^+| \leq |\varepsilon_{2d}||\varepsilon^+| < |\varepsilon_{2d}|\, |\varepsilon_{2d+1}|$. □

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8 The Optimal Weight Formula

A key question is: for which integer $w \geq 1$ do the two consecutive weighted mediants

$$r_- = \text{Mediant}(r_{2d}, r_{2d+1}, w-1), \qquad r_+ = \text{Mediant}(r_{2d}, r_{2d+1}, w)$$

straddle θ, i.e. satisfy the bracketing condition $(\lambda_- - 1)(\lambda_+ - 1) < 0$?

Writing $A = e_{2d} - \theta\, o_{2d}$ and $B = e_{2d+1} - \theta\, o_{2d+1}$ for the scaled errors, the bracketing condition becomes $(A + (w-1)B)(A + wB) < 0$. Observe that $A + (w-1)B$ and $A + wB$ are consecutive evaluations of the linear function $f(w) = A + wB$ at $w - 1$ and w: their product is negative iff f changes sign between $w - 1$ and w, i.e. iff the zero $-A/B$ of f lies in the interval $(w-1, w)$, i.e. iff $w = \lfloor -A/B \rfloor + 1$. Since θ is irrational, $-A/B$ is never an integer, so exactly one such w exists. The unique integer satisfying this bracketing condition is given by the following closed-form expression.

Theorem 8.1 (Optimal weight)**.** *Let (r_{2d}, r_{2d+1}) be a bounded pair bracketing θ, with*

$$A = e_{2d} - \theta\, o_{2d}, \qquad B = e_{2d+1} - \theta\, o_{2d+1},$$

and set $\Delta = e_{2d+1}o_{2d} - e_{2d}o_{2d+1}$, $\Theta = o_{2d+1}B$. *Then the unique integer* $w \geq 1$ *for which* $r_- = \text{Mediant}(r_{2d}, r_{2d+1}, w-1)$ *and* $r_+ = \text{Mediant}(r_{2d}, r_{2d+1}, w)$ *satisfy the bracketing condition* $(\lambda_- - 1)(\lambda_+ - 1) < 0$ *is*

$$w^+ = \left\lfloor \frac{-A}{B} \right\rfloor + 1 = \left\lfloor \frac{\Delta}{\Theta} - \frac{o_{2d}}{o_{2d+1}} \right\rfloor + 1.$$

A derivation is given in Appendix A.

When $\Theta = 0$, r_{2d+1} is an exact rational representation of θ and no further improvement is possible; the algorithm terminates.

Corollary 8.2 (Children at w^+ form a bounded pair)**.** *Under the conditions of Theorem 8.1, the two children* $r_- = \text{Mediant}(r_{2d}, r_{2d+1}, w^-)$ *and* $r_+ = \text{Mediant}(r_{2d}, r_{2d+1}, w^+)$ *form a bounded pair.*

Proof. Farey adjacency $|e_+o_- - e_-o_+| = 1$ follows from Theorem 7.1. It remains to show both children are good estimates. Write $f(w) = A + wB$ so that $f(w) = e_w - \theta o_w$ (the scaled error of r_w). The bracketing condition $f(w^-)\, f(w^+) < 0$ means the two values have opposite signs, so

$$|f(w^-)|\, o_+ + |f(w^+)|\, o_- = |f(w^-)\, o_+ - f(w^+)\, o_-| = |e_-\, o_+ - e_+\, o_-| = |\Delta| = 1.$$

Both terms on the left are in fact positive: since θ is irrational (Remark 6.1), $f(w) \neq 0$ for all integers w, so $|f(w^-)|$ and $|f(w^+)|$ are both nonzero; and $o_-, o_+ \geq 1$. Since two positive terms sum to 1, each is strictly less than 1, i.e. $|e_- - \theta o_-| < 1$ and $|e_+ - \theta o_+| < 1$, confirming both children are good estimates. Moreover, $|\Delta| = 1$ directly gives $\gcd(e_+, o_+) = \gcd(e_-, o_-) = 1$, so the children are ratios in lowest terms. □

DRAFT

The continuous optimum and the discrete formula coincide

To see why w^+ is the natural choice, consider treating w as a real variable and finding the stationary point of the error product $f(w-1)\, f(w)$, where $f(w) = A + wB$. Differentiating with respect to w and setting the derivative to zero:

$$\frac{d}{dw}\big[f(w-1)\, f(w)\big] = B\big[(A + wB) + (A + (w-1)B)\big] = 0.$$

Solving gives the unique real critical point

$$w^* = \tfrac{1}{2} - \frac{A}{B} = c + \tfrac{1}{2},$$

where $c = -A/B > 0$. The product $f(w-1)\, f(w)$ has roots at $w = c$ (where $f(w) = 0$) and $w = c + 1$ (where $f(w-1) = 0$), and $w^* = c + \frac{1}{2}$ is their midpoint — the unique real maximum of $|f(w-1)\, f(w)|$ between the roots. The unique integer w for which $f(w-1)$ and $f(w)$ have opposite signs satisfies $c < w < c+1$, i.e. $w = \lfloor c \rfloor + 1 = w^+$. Since $w^* = c + \frac{1}{2}$ also lies in $(c,\, c+1)$, and w^+ is the unique integer in that interval, w^+ is the nearest integer to w^*: both the continuous stationary point and the sign-change bracket single out the same w^+.

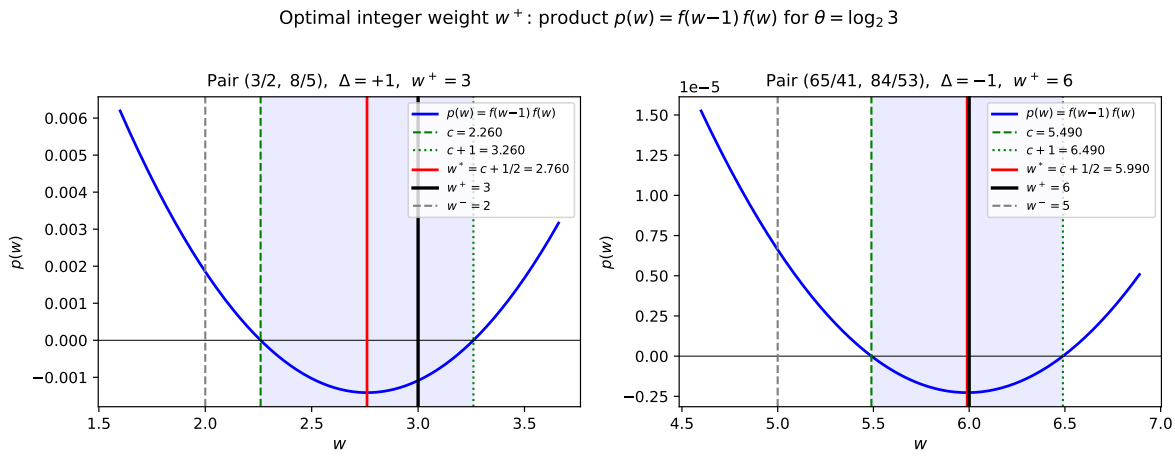


Figure 1: The product $p(w) = f(w-1)\,f(w)$ for two pairs from the mediant tree for $\theta = \log_2 3$, illustrating the optimal integer weight w^+. *Left:* pair $(3/2,\ 8/5)$ with $\Delta = +1$, $c \approx 2.260$, and $w^+ = 3$. *Right:* pair $(65/41,\ 84/53)$ with $\Delta = -1$, $c \approx 5.490$, and $w^+ = 6$. In each case the roots of p are at c (where $f(w) = 0$, green dashed) and $c+1$ (where $f(w-1) = 0$, green dotted); the continuous stationary point $w^* = c + \frac{1}{2}$ (red) is their midpoint; the unique integer in the shaded interval $(c,\ c+1)$ is w^+ (black); and its companion $w^- = w^+ - 1$ (grey dashed) lies just outside the interval on the left. Together w^- and w^+ are the two weights whose mediants bracket θ. The two panels illustrate both signs of Δ.

DRAFT

Algorithm. The iteration proceeds as follows.

1. Start with an initial bounded pair (r_0, r_1) for θ.
2. Compute Δ and Θ.
3. If $\Theta = 0$, terminate.
4. Compute $w^+ = \lfloor \Delta/\Theta - o_{2d}/o_{2d+1} \rfloor + 1$ and set $w^- = w^+ - 1$.
5. Form the two mediants $\mathrm{Mediant}(r_{2d}, r_{2d+1}, w^-)$ and $\mathrm{Mediant}(r_{2d}, r_{2d+1}, w^+)$, order them so that r_{2d+3} is the better estimate (smaller $|\varepsilon|$) and r_{2d+2} the worse, and repeat from step 2.

When $w^+ \geq 2$, the denominator of each new mediant strictly exceeds both parent denominators. When $w^+ = 1$ only one new fraction r^+ is produced (the non-crossing "child" r^- coincides with r_{2d}), but r^+ has strictly larger denominator. In either case the denominators of the better estimates grow without bound and the iteration converges to θ (Corollary 10.2).

9 Mediant Words and the Mediant Tree

The mediant construction admits a compact combinatorial encoding. Every estimate produced by the algorithm is either a *leaf* or a *branch*, and the recursive structure of all estimates defines a binary tree whose nodes carry both rational values and combinatorial words over $\{\mathsf{O}, \mathsf{E}\}$.

9.1 Leaf words

Definition 9.1 (Leaf mediant word)**.** Let $\{\mathsf{O},\mathsf{E}\}^*$ denote the set of finite strings over the two-letter alphabet $\{\mathsf{O},\mathsf{E}\}$. A *leaf* estimate has denominator $o = 1$, i.e. it is a base estimate $e/1$ for some positive integer e. Its *mediant word* is

$$\pi(e/1) = \underbrace{\mathsf{E}\cdots\mathsf{E}}_{e}\mathsf{O},$$

a string of e copies of E followed by one O.

The two leaf estimates that initialise the algorithm are $\lfloor\theta\rfloor/1$ and $\lceil\theta\rceil/1$, ordered as (r_0, r_1) so that r_1 is the better estimate (i.e. $|\varepsilon_{r_1}| < |\varepsilon_{r_0}|$). Their mediant words are $\mathsf{E}^{\lfloor\theta\rfloor}\mathsf{O}$ and $\mathsf{E}^{\lceil\theta\rceil}\mathsf{O}$ respectively. Since θ is irrational, $\lceil\theta\rceil = \lfloor\theta\rfloor + 1$, so $\pi(r_0)$ and $\pi(r_1)$ differ by exactly one E. Each word $\mathsf{E}^k\mathsf{O}$ contains exactly one O, so it is not a power of a shorter word; both $\pi(r_0)$ and $\pi(r_1)$ are therefore primitive. Moreover, $\mathsf{E}^k\mathsf{O}$ is the lower Christoffel word $\mathrm{Ch}^-(k, 1)$ (Definition 5.1): its O appears at position k, matching the Christoffel formula. Thus both leaf words are lower Christoffel words of their respective fractions.

9.2 Branch words and the weighted operators

Each branch step produces two new mediant words from the parent pair $(u, v) = (\pi(r_{2d-2}), \pi(r_{2d-1}))$ by concatenation:

Definition 9.2 (Branch mediant word)**.** The bounded pair at depth d consists of estimates r_{2d} and r_{2d+1}, with r_{2d+1} always the better estimate (smaller $|\varepsilon|$). Each estimate r_{2d} or r_{2d+1} has two *ordered parents* (r_{2d-2}, r_{2d-1}) and a non-negative integer *weight* w (equal to zero only when $w^+_{d-1} = 1$, in which case $r_{2d} = r_{2d-2}$). The two estimates r_{2d} and r_{2d+1} carry weights w^-_{d-1} and w^+_{d-1}. We distinguish two uses of the word *weight*:

- The *inherited weights* w_{2d} and w_{2d+1} of the two estimates r_{2d} and r_{2d+1} are the weights with which they were formed from the pair (r_{2d-2}, r_{2d-1}) at depth $d-1$ — specifically $\{w_{2d}, w_{2d+1}\} = \{w^-_{d-1}, w^+_{d-1}\}$ with $w_{2d} = w^+_{d-1} - 1$ and $w_{2d+1} = w^+_{d-1}$ (or vice versa, depending on which child is the better estimate). Each inherited weight is used to form exactly one of the two estimates: $r_{2d} = \mathrm{Mediant}(r_{2d-2}, r_{2d-1}, w_{2d})$ and $r_{2d+1} = \mathrm{Mediant}(r_{2d-2}, r_{2d-1}, w_{2d+1})$.

- The *propagated weights* w^-_d and w^+_d of a bounded pair (r_{2d}, r_{2d+1}) are the optimal weights computed *from* that pair to produce the next pair (r_{2d+2}, r_{2d+3}) — i.e. the w^- and w^+ of Theorem 8.1 applied to (r_{2d}, r_{2d+1}).

These differ in general: the inherited weights w_{2d} and w_{2d+1} record how r_{2d} and r_{2d+1} were born; the propagated weights w^-_d and w^+_d determine where the algorithm goes next. Write $u = \pi(r_{2d-2})$ and $v = \pi(r_{2d-1})$ for the parent mediant words at depth $d-1$.

The two branch words at each depth share the same parents and differ only in the multiplicity of v: one estimate uses weight w^+_{d-1} and the other uses weight w^-_{d-1}. Which

DRAFT

is the better estimate depends on the errors: writing ε^- and ε^+ for the errors of the two candidates (Definition 6.6),

$$w_{2d+1} - w_{2d} = \begin{cases} +1 & \text{if } |\varepsilon^+| < |\varepsilon^-|, \\ -1 & \text{otherwise.} \end{cases}$$

In either case $|\pi(r_{2d+1})| - |\pi(r_{2d})| = \pm|\pi(r_{2d-1})|$.

The branch mediant words are (using $w = w^+_{d-1}$, the propagated optimal weight of the parent pair (r_{2d-2}, r_{2d-1}), and Δ the determinant of that parent pair):

- $\Delta > 0$ (r_{2d-1} is the upper bound, v plays the role of E):

$$\pi(r^-) = v^{w^+_{d-1}-1} \cdot u,$$
$$\pi(r^+) = v^{w^+_{d-1}} \cdot u.$$

- $\Delta < 0$ (r_{2d-1} is the lower bound, v plays the role of O):

$$\pi(r^-) = u \cdot v^{w^+_{d-1}-1},$$
$$\pi(r^+) = u \cdot v^{w^+_{d-1}}.$$

Which of r^- and r^+ is the better estimate r_{2d+1} (and which is r_{2d}) is determined by comparing $|\varepsilon^-|$ and $|\varepsilon^+|$; the assignment is not fixed in advance.

DRAFT

Concrete example (for $\theta = \log_2 3$: $w = 2$, depth-0 leaves $\pi(r_0) = \mathtt{EO}$, $\pi(r_1) = \mathtt{EEO}$, $\Delta > 0$). Write $\Phi^w(v, u) = (v^w \cdot u,\ v^{w-1} \cdot u)$ for the pair of branch words at weight w (the $\Delta > 0$ case):

$$\Phi^2\big(\mathtt{EEO},\, \mathtt{EO}\big) = \big(\mathtt{EEO}^2\,\mathtt{EO},\ \mathtt{EEO}\,\mathtt{EO}\big) = \big(\mathtt{EEOEEOEO},\ \mathtt{EEOEO}\big),$$

so $\pi(r_3) = \mathtt{EEOEEOEO}$ and $\pi(r_2) = \mathtt{EEOEO}$ (the second component of $\Phi^2(\mathtt{EEO}, \mathtt{EO})$, i.e. the weight-$w^- = 1$ child).

9.3 Mediant words are Christoffel words

A word is *primitive* if it is not a power x^k of a strictly shorter word x (with $k \geq 2$).

Lemma 9.3 (Every mediant word is a lower Christoffel word)**.** *For every estimate $r = (e, o)$ produced by the mediant tree, $\pi(r) = \mathrm{Ch}^-(e, o)$, the lower Christoffel word at slope $o/(e+o)$. In particular, $\pi(r)$ is never an upper Christoffel word, since by Theorem 12.1 every mediant word ends in O, whereas upper Christoffel words end in E.*

Proof. By structural induction on the mediant tree.

Base case. By §9.1, $\pi(e/1) = \mathsf{E}^e\mathsf{O} = \mathrm{Ch}^-(e, 1)$.

Inductive step. Let (r_{2d}, r_{2d+1}) be a bounded pair with $\Delta = e_{2d+1}o_{2d} - e_{2d}o_{2d+1}$. By induction $\pi(r_{2d}) = \mathrm{Ch}^-(e_{2d}, o_{2d})$ and $\pi(r_{2d+1}) = \mathrm{Ch}^-(e_{2d+1}, o_{2d+1})$, both lower Christoffel words (ending in O), with slopes o_{2d}/L_{2d} and o_{2d+1}/L_{2d+1} (where $L_k = e_k + o_k$) on opposite sides of $\alpha = 1/(1+\theta)$.

When $\Delta > 0$: r_{2d+1} is the upper estimate so $e_{2d+1}/o_{2d+1} > \theta$, giving $o_{2d+1}/L_{2d+1} < \alpha < o_{2d}/L_{2d}$. The new word is $v^w u = \pi(r_{2d+1})^w \cdot \pi(r_{2d})$: the smaller-slope word $\pi(r_{2d+1})$ repeated w times, followed by $\pi(r_{2d})$ once. By Theorem 5.2 (with $e_0 = e_{2d+1}$, $e_1 = e_{2d}$) this equals $\mathrm{Ch}^-(w\,e_{2d+1} + e_{2d},\ w\,o_{2d+1} + o_{2d})$.

When $\Delta < 0$: r_{2d+1} is the lower estimate so $e_{2d+1}/o_{2d+1} < \theta$, giving $o_{2d}/L_{2d} < \alpha < o_{2d+1}/L_{2d+1}$. Thus $\pi(r_{2d})$ has the smaller slope and $\pi(r_{2d+1})$ has the larger slope. The new word is $uv^w = \pi(r_{2d}) \cdot \pi(r_{2d+1})^w$: the single smaller-slope word followed by the larger-slope word repeated w times. By Theorem 5.2 (with $e_0 = e_{2d}$, $e_1 = e_{2d+1}$, $o_0/L_0 < o_1/L_1$, Berstel [5] Theorem 1.5.1) this equals $\mathrm{Ch}^-(e_{2d} + w\,e_{2d+1},\ o_{2d} + w\,o_{2d+1})$.

In both cases $\gcd(e_{2d} + we_{2d+1},\ o_{2d} + wo_{2d+1}) = 1$ by the Farey adjacency condition $|e_{2d+1}o_{2d} - e_{2d}o_{2d+1}| = 1$, confirming the result is a valid lower Christoffel word. □

The Christoffel identification is the key fact connecting the mediant algorithm to the combinatorics of Sturmian words. The recurrence $\pi(r_{2d+3}) = v^{w_d^\pm} \cdot u$ (or $u \cdot v^{w_d^\pm}$), where $u = \pi(r_{2d})$ and $v = \pi(r_{2d+1})$ are the parent words at depth d and $w_d^\pm$ is the propagated weight of the pair (r_{2d}, r_{2d+1}), matches the standard sequence generation rule for Sturmian words (Widmer [4], Proposition 2.4.6).

Remark 9.4. The Christoffel tree — the binary tree whose nodes are lower Christoffel words under the same concatenation rules — is a classical object (Berstel et al. [5], Chapter 2). Lemma 9.3 says that the mediant algorithm traverses exactly this tree: every estimate produced is a node of the Christoffel tree, and the branch rule is the Christoffel concatenation. The novelty is not the tree itself but the *driving criterion*: the optimal weight w^+ (Theorem 8.1) selects the unique child that straddles θ, making the traversal goal-directed rather than exhaustive.

DRAFT

9.4 The mediant tree

The mediant tree is the practical device for generating mediant words of arbitrary length and precision. Given a target θ and a desired precision, one runs the algorithm until the bounded pair (r_{2d}, r_{2d+1}) achieves the required length $e_{2d+1} + o_{2d+1}$. The mediant word $\pi(r_{2d+1})$ is then read off the tree by the recursive rule: an interior node $\mathrm{Mediant}(s, t, w)$ with $\Delta > 0$ yields $\pi(t)^w \cdot \pi(s)$, and with $\Delta < 0$ yields $\pi(s) \cdot \pi(t)^w$. Deeper trees produce longer prefixes of the Sturmian word with slope $1/(\theta + 1)$ (Section 12).

Definition 9.5 (Mediant tree)**.** The *mediant tree* of an estimate r is defined recursively:

- If $r \in \{\lfloor\theta\rfloor, \lceil\theta\rceil\}$, then r is a *leaf node*.
- Otherwise r is an *interior node* $\mathrm{Mediant}(s,\ t,\ w)$ with left subtree s, right subtree t, and integer weight $w \geq 0$, representing the rational

$$r = \frac{e_s + w\,e_t}{o_s + w\,o_t}.$$

The two leaves are the unique consecutive integers bracketing θ, forming the initial bounded pair (r_0, r_1), ordered so that r_1 is the better estimate (i.e. $|\varepsilon_{r_1}| < |\varepsilon_{r_0}|$). Each

bounded pair (r_{2d}, r_{2d+1}) in the algorithm's sequence consists of two mediant trees sharing the same leaves; as the iteration advances the trees grow deeper, recording the full construction history of each estimate.

A detailed worked example for $\theta = \log_2 3$, tracing the first four steps of the algorithm and the corresponding mediant word decompositions, is given in Appendix B.

9.5 The mediant word encodes the fraction

By Lemma 9.3, the mediant word $\pi(r)$ of an estimate $r = e/o$ is uniquely determined by its fraction: $\pi(r) = \mathrm{Ch}^-(e, o)$. The fraction e/o records the full construction history of r via the Euclidean structure of the mediant algorithm. Conversely, the fraction e/o can be read off from $\pi(r)$ by counting E's and O's. As the iteration advances, the ratio of E's to O's in $\pi(r_{2d+1})$ converges to θ, with strict improvement guaranteed across iterations (Theorem 10.1) but not within a single depth (the worse estimate r_{2d} need not improve).

10 Convergence Analysis

Theorem 10.1 (Strict decrease of $|\varepsilon|$)**.** *At each iteration of the algorithm, $|\varepsilon_{2d+1}|$ strictly decreases.*

Proof. By Theorem 7.1, $|\varepsilon^+| < |\varepsilon_{2d+1}|$. The next iterate assigns the better of r_- and r_+ as r_{2d+3}, so $|\varepsilon_{2d+3}| \le |\varepsilon^+| < |\varepsilon_{2d+1}|$. □

DRAFT

Corollary 10.2 (Convergence to θ)**.** *The sequence of better estimates produced by the algorithm satisfies $\varepsilon_{2d+1} \to 0$.*

Proof. From the proof of Corollary 8.2, writing $f(w) = A + wB$ for the linear error function and o_+, o_- for the two child denominators:

$$|f(w^-)|\, o_+ + |f(w^+)|\, o_- = |\Delta| = 1.$$

Since $f(w) = e_w - \theta\, o_w$ and $\varepsilon_w = f(w)/(\theta\, o_w)$, substituting $|f(w)| = |\varepsilon_w|\, \theta\, o_w$ gives

$$\big(|\varepsilon^-| + |\varepsilon^+|\big)\, \theta\, o_-\, o_+ = 1,$$

so in particular the better child satisfies

$$|\varepsilon_{\text{better}}| \le \frac{1}{2\,\theta\, o_-\, o_+}.$$

(When $w^+ = 1$, so $w^- = 0$: $r^- = r_{2d}$ coincides with the parent and is not a new fraction; the better child is r^+ in this case since $|\varepsilon^+| < |\varepsilon_{2d+1}| < |\varepsilon_{2d}|$. If the better child is r^-, then $w^- \ge 1$ and $o^- = o_{2d} + (w^+ - 1)\, o_{2d+1} > o_{2d+1}$. In either case the bound $|\varepsilon_{\text{better}}| \le 1/(2\theta\, o_-\, o_+)$ holds.) Since $o_+ = o_{2d} + w^+\, o_{2d+1} \ge o_{2d} + o_{2d+1} > o_{2d+1}$, the denominator of the better child strictly exceeds o_{2d+1}; by induction the denominators diverge. Since $|\varepsilon_{\text{better}}| \le 1/(2\,\theta\, o_-\, o_+)$ and $o_+ \to \infty$, we have $\varepsilon_{2d+1} \to 0$. □

Connection to continued fractions

The connection between the mediant algorithm and continued fractions is a direct consequence of the Sturmian identity (Theorem 12.6). In combinatorics on words, the standard sequence for a Sturmian word with slope $\alpha = 1/(1+\theta)$ is generated by the recurrence $s_n = s_{n-1}^{b_n} \cdot s_{n-2}$, where the weights b_n are precisely the partial quotients of the continued-fraction expansion of α (Lothaire [3], Ch. 2; Widmer [4], Proposition 2.4.6); here $b_1 = a_0$ and $b_n = a_{n-1}$ for $n \geq 2$, with $\alpha = 1/(1+\theta) = [0;\, a_0+1,\, a_1,\, a_2,\, \ldots]$. The mediant word sequence agrees with this standard sequence at all interior positions (Theorem 12.6). This does *not* imply that the algorithm's weights w^+ equal the partial quotients b_k: the Sturmian identity constrains only interior positions of the words, not which weights the algorithm selects. What does follow is that the two children at weights w^- and w^+ always have denominators $o_{2d} + w^- \cdot o_{2d+1}$ and $o_{2d} + w^+ \cdot o_{2d+1}$ (one even-indexed, one odd-indexed depending on which is the better estimate), mirroring the structure of the classical recurrence $q_{k+1} = b_k q_k + q_{k-1}$. Exponential denominator growth is proved in Section 10.

The MT fractions are not in general the same as the CF convergents. The standard CF recurrence jumps by the full partial quotient a_k at each step, landing on a convergent p_k/q_k. The MT algorithm picks the optimal bracketing weight w^+, which need not equal any partial quotient.

Theorem 10.3 (MT fractions lie on the Stern–Brocot path)**.** *Every fraction appearing as r_{2d} or r_{2d+1} in a bounded MT pair is a convergent or semi-convergent of θ, i.e. lies on the Stern–Brocot path of θ.*

DRAFT

Proof. The initial bounds $\lfloor\theta\rfloor/1$ and $\lceil\theta\rceil/1$ are consecutive integers, hence Farey adjacent, and both lie on the Stern–Brocot path of θ.

At each depth d, the algorithm forms the two mediants $r^{\pm} = \mathrm{Mediant}(r_{2d}, r_{2d+1}, w^{\pm})$ with $w^- = w^+ - 1$. For $w \geq 2$ these are *weighted* mediants, not ordinary mediants of (r_{2d}, r_{2d+1}), so the one-step argument does not apply directly. Instead, use the identity

$$P + wQ = (P + (w-1)Q) \oplus Q, \qquad \det(P + (w-1)Q,\, Q) = \det(P, Q),$$

which shows that $\mathrm{Mediant}(r_{2d}, r_{2d+1}, w)$ is the ordinary mediant of $(r_{2d} + (w-1)\, r_{2d+1},\ r_{2d+1})$, a Farey-adjacent pair with the same determinant as (r_{2d}, r_{2d+1}). By induction on $w = 1, \ldots, w^+$: the pair $(r_{2d} + (w-1)\, r_{2d+1},\ r_{2d+1})$ is Farey adjacent and straddles θ for each $w \leq w^+$. By the standard Stern–Brocot lemma, any Farey-adjacent pair (P, Q) of non-negative fractions defines a unique SB-tree interval, and θ lies in the interval of (P, Q) if and only if $P \oplus Q$ lies on the SB path to θ; so its ordinary mediant $r_{2d} + w\, r_{2d+1}$ is a node on the Stern–Brocot path of θ. The case $w^- = 0$ (i.e. $w^+ = 1$) is immediate: $r^- = r_{2d}$ is already on the path by inductive hypothesis, and $r^+ = r_{2d} + r_{2d+1}$ is the unique ordinary mediant of the Farey pair (r_{2d}, r_{2d+1}). Both r^- and r^+ are produced this way, so both lie on the path. □

Theorem 10.4 (MT never skips a convergent)**.** *Every CF convergent p_k/q_k of θ appears as r_{2d} or r_{2d+1} in some bounded MT pair.*

Proof. A convergent p_k/q_k is characterised as the *last* node in a run of same-side steps on the SB path of θ, i.e. the node just before the path crosses to the other side. In the induction

of Theorem 10.3, the path nodes visited within stage d are $r_{2d} + w\, r_{2d+1}$ for $w = 1, \ldots, w^+$. For $w < w^+$ the node lies on the same side of θ as r_{2d}. When $w^+ \geq 2$ the non-crossing child $r^- = r_{2d} + (w^+ - 1)\, r_{2d+1}$ is the last node on r_{2d}'s side, so r^- is a convergent. (When $w^+ = 1$ there is no non-crossing child: r_{2d} itself is carried forward and it is a convergent from the previous stage.)

We show that every convergent appears in the MT sequence. After the integer spine and the two leaves, the Stern–Brocot path of θ is the concatenation of the stage blocks $\{r_{2d} + w\, r_{2d+1} : 1 \leq w \leq w_d^+\}$, $d = 0, 1, 2, \ldots$. (After the last node r^+ of block $d-1$, the SB interval is (r_{2d}, r_{2d+1}), so the next path node is $r_{2d} \oplus r_{2d+1}$, the first node of block d.) Within a block, the nodes with $w < w_d^+$ lie on the same side of θ as r_{2d}, and r^+ lies on the other side. The nodes that end a run—and hence are convergents—are: $r^- = r_{2d} + (w_d^+ - 1)\, r_{2d+1}$ (when $w_d^+ \geq 2$); r^+; and, when $w_d^+ = 1$, r_{2d} itself (carried forward from the previous block). All of these are members of an MT pair. Together with the leaf $\lfloor\theta\rfloor/1 = p_0/q_0$, every convergent appears. □

The next two results give precise characterisations of which MT fractions are semi-convergents rather than convergents.

Theorem 10.5 (Semi-convergent characterisation of r_{2d+1})**.** *For all $d \geq 0$, r_{2d+1} is a semi-convergent of θ if and only if $w_d^+ = 1$, i.e. the propagated weight of the pair (r_{2d}, r_{2d+1}) equals 1.*

Proof. Write $A = e_{2d} - \theta o_{2d}$ and $B = e_{2d+1} - \theta o_{2d+1}$ for the signed errors of the current pair, so $w_d^+ = \lfloor |A|/|B| \rfloor + 1$.

Case $d \geq 1$**.** Let r^-, r^+ be the children formed at depth $d-1$, with $o^- < o^+$.

Case $r_{2d+1} = r^+$ (the crossing child). After the path visits r^+, the SB interval is $(r^-, r^+) = (r_{2d}, r_{2d+1})$, so the next path node is $r_{2d} \oplus r_{2d+1}$. By definition of $w_d^+ = \lfloor |A|/|B| \rfloor + 1$, $w_d^+ = 1$ iff $|A| < |B|$, i.e. $r_{2d} \oplus r_{2d+1}$ lies on the same side as r_{2d+1}, i.e. the path crosses immediately. Hence r^+ is the last node of its run (a convergent) iff $w_d^+ \geq 2$, and a semi-convergent iff $w_d^+ = 1$.

Case $r_{2d+1} = r^-$ (the non-crossing child). We have $w_{d-1}^- \geq 1$ (otherwise $r^- = r_{2d-2}$ would be the worse child, contradicting $|\varepsilon_{2d-2}| > |\varepsilon_{2d-1}|$), so the path visits r^- immediately before r^+, which lies on the other side. Hence r^- is a convergent. Also $o_{2d+1} = o^- < o^+ = o_{2d}$, so $|A| \geq |B|$ and $w_d^+ \geq 2$. The statement holds in this case too.

Case $d = 0$**.** The seed pair is $(r_0, r_1) = (\lfloor\theta\rfloor/1, \lceil\theta\rceil/1)$ or $(\lceil\theta\rceil/1, \lfloor\theta\rfloor/1)$, ordered so that r_1 is the better estimate (smaller $|\varepsilon|$). Write $\alpha = \{\theta\} \in (0, 1)$. Then $|A| = |\varepsilon_{r_0}|$ and $|B| = |\varepsilon_{r_1}|$ are $\{1 - \alpha, \alpha\}$ in some order, with $|B| < |A|$ by definition of the ordering. Hence $|A|/|B| > 1$, so $w_0^+ = \lfloor |A|/|B| \rfloor + 1 \geq 2$. Concretely: if $r_1 = \lfloor\theta\rfloor/1$ (i.e. $\alpha < \frac{1}{2}$) then $|A| = 1 - \alpha$, $|B| = \alpha$, and $|A|/|B| = (1-\alpha)/\alpha > 1$; if $r_1 = \lceil\theta\rceil/1$ (i.e. $\alpha > \frac{1}{2}$) then $|A| = \alpha$, $|B| = 1 - \alpha$, and $|A|/|B| = \alpha/(1-\alpha) > 1$. Since $w_0^+ \geq 2 \neq 1$, the biconditional requires r_1 not to be a semi-convergent when $w_0^+ \neq 1$. By Remark 10.8, if $\lceil\theta\rceil/1$ is the better leaf then $\{\theta\} > \frac{1}{2}$, so $a_1 = 1$ and $\lceil\theta\rceil/1 = p_1/q_1$ is the first convergent; and $\lfloor\theta\rfloor/1 = p_0/q_0$ is always a convergent. Hence r_1 is a convergent (not a semi-convergent) at $d = 0$, and the statement holds.

In all cases, r_{2d+1} is not on the integer spine ($o_{2d+1} \geq 2$ for $d \geq 1$; the leaves are treated separately), so the characterisation is complete. □

DRAFT

Remark 10.6. Note that w_d^+ (the propagated weight of (r_{2d}, r_{2d+1})) differs in general from w_{2d+1} (the inherited weight of r_{2d+1}). A statement in terms of the inherited weight w_{2d+1} is false: for $\theta = \log_2 3$, $r_5 = 8/5$ carries inherited weight 1 but is a convergent, and $r_3 = 5/3$ carries inherited weight 2 but is a semi-convergent.

Theorem 10.7 (Semi-convergent characterisation of r_{2d})**.** *For $d \geq 1$, r_{2d} is a semi-convergent of θ if and only if $w_{2d} > w_{2d+1}$, i.e. the inherited weight of r_{2d} exceeds that of its sibling r_{2d+1}.*

Proof. Let r^-, r^+ be the children formed at depth $d-1$, with absolute errors $x = |e^- - \theta o^-|$ and $y = |e^+ - \theta o^+|$ and $o^- < o^+$.

($\Leftarrow$) Suppose $w_{2d} > w_{2d+1}$, so $r_{2d} = r^+$ is the crossing (worse) child. Then $x/o^- < y/o^+$. Since $o^- < o^+$, this gives $x < y$.

Since (r^-, r^+) is a bounded pair (Theorem 7.1), the pair brackets θ: their signed errors $e^- - \theta o^-$ and $e^+ - \theta o^+$ have opposite signs. Write them as $-x$ and $+y$ respectively (i.e. r^- is the underestimate and r^+ the overestimate; the argument is symmetric if the signs are reversed). The signed error of the unweighted mediant $r^- \oplus r^+$ is then

$$(e^- + e^+) - \theta(o^- + o^+) = (-x) + y = y - x > 0,$$

which has the same sign as the error of r^+. Hence θ still lies on the r^- side of $r^- \oplus r^+$, so the SB path continues toward r^+: the crossing has not yet occurred, and r^+ is not the last node of its run. Hence r_{2d} is a semi-convergent.

($\Rightarrow$) Suppose $w_{2d} \leq w_{2d+1}$, so $r_{2d} = r^-$ is the non-crossing child, i.e. $r^- = r_{2d-2} + (w_{d-1}^+ - 1)\, r_{2d-1}$.

- If $w_{d-1}^- \geq 1$: by the block structure established in the proof of Theorem 10.3, the SB path visits the nodes $r_{2d-2} + w\, r_{2d-1}$ for $w = 1, \ldots, w_{d-1}^+$ in order within stage $d-1$. All nodes with $w < w_{d-1}^+$ lie on the same side of θ as r_{2d-2}; the node at $w = w_{d-1}^+$ (namely $r_{2d-2} + w_{d-1}^+ r_{2d-1} = r_{2d+1}$) lies on the other side. Since $r^- = r_{2d-2} + (w_{d-1}^+ - 1)\, r_{2d-1}$ is the node at $w = w_{d-1}^+ - 1 \geq 1$, it is the last same-side node before the path crosses to r^+. Hence r^- is the last node of its run, i.e. a convergent.

- If $w_{d-1}^- = 0$: then $r_{2d} = r_{2d-2}$ and $w_{d-1}^+ = 1$. By the induction hypothesis ($\Rightarrow$) at depth $d-1$: since $w_{2d-2} \leq w_{2d-1}$ (as r_{2d-2} is the non-crossing child at depth $d-1$), r_{2d-2} is a convergent. Hence $r_{2d} = r_{2d-2}$ is a convergent. (The base case $d = 1$: $w_0^+ \geq 2$ always, so $w_0^- = 0$ forces $w_0^+ = 1$, a contradiction; this case cannot occur.)

□

Remark 10.8. $\lfloor \theta \rfloor / 1 = p_0/q_0$ is always a convergent. $\lceil \theta \rceil / 1$ is a convergent ($= p_1/q_1$) iff $a_1 = 1$, and otherwise a semi-convergent. If $\lceil \theta \rceil / 1$ is the better leaf, then $\{\theta\} > 1/2$, so $a_1 = 1$; hence r_1 is always a convergent, and r_0 is a semi-convergent iff $r_0 = \lceil \theta \rceil / 1$ and $a_1 > 1$. Leaves carry no weight, so the weight condition does not apply at $d = 0$.

All steps in both proofs are argued by hand.

DRAFT

Theorem 10.9 (Linear depth bound)**.** *Let $\theta > 1$ be irrational. For every depth d and both members e_d/o_d of the depth-d pair,*

$$d \;\le\; 2\log_2(e_d + o_d) + 1.$$

In particular, the depth grows at most logarithmically in $e_d + o_d$, so $O(\log N)$ iterations of the mediant-tree algorithm suffice to produce an approximant of size $N = e_d + o_d$.

Proof. We establish two lemmas and then combine them.

Lemma A (no two consecutive unit weights). $w_0^+ \ge 2$ always (the depth-0 weight is at least 2), and for no d does $w_d^+ = w_{d+1}^+ = 1$ hold simultaneously.

Proof of Lemma A. Let $A = e_0 - \theta o_0$ and $B = e_1 - \theta o_1$ be the signed errors of the two current estimates, with $|B| < |A|$ (so the better estimate is e_1/o_1). The weight $w_d^+ = \lfloor -A/B \rfloor + 1$; in particular $w_d^+ = 1$ iff $|A| < 2|B|$. Since $|B|/o_1 \le |A|/o_0$ (the better estimate has smaller error per denominator), $w_0^+ \ge 2$ at $d = 0$.

Suppose $w_d^+ = k \ge 2$, so the two children have errors $s = |A| - (k-1)|B|$ and $t = k|B| - |A|$ with denominators $o^- = o_0 + (k-1)o_1 < o^+ = o_0 + ko_1$. If the worse child r^- becomes the new coarse bound, then $s/o^- < t/o^+$, giving $s < t$, so $w_{d+1}^+ \ge 2$. If the better child r^+ becomes the new coarse bound, $w_{d+1}^+ = 1$ means $s < t$. For $w_{d+2}^+ = 1$ we would then need $2s < t$. But $t < s \cdot o^+/o^- \le 2s$ because $o^+ \le 2o^-$ whenever $k \ge 2$. Hence $w_{d+2}^+ \ge 2$. In either case two consecutive unit weights are impossible. $\square_A$

Lemma B (denominator growth). For every depth d, both denominators o_d satisfy $o_d \ge 2^{\lfloor d/2 \rfloor}$.

Proof of Lemma B. By induction. The base cases $d = 0, 1$ hold since $o_0, o_1 \ge 1$. For the inductive step, every child denominator is at least o_0 (it is $o_0 + wo_1 \ge o_0$). When $w_d^+ \ge 2$, both children have denominator $\ge o_0 + o_1$. By Lemma A, among any two consecutive depths at least one has weight ≥ 2, so every pair of consecutive denominators satisfies $o_{d+1} \ge o_d + o_{d-1} \ge 2\min(o_d, o_{d-1})$. Unrolling the induction gives $o_d \ge 2^{\lfloor d/2 \rfloor}$. $\square_B$

Conclusion. From Lemma B, $o_d \ge 2^{\lfloor d/2 \rfloor}$, so $\lfloor d/2 \rfloor \le \log_2 o_d \le \log_2(e_d + o_d)$. Hence $d \le 2\log_2(e_d + o_d) + 1$. $\square$

The constants $(m_\theta, b_\theta) = (2, 1)$ are universal. The optimal per-target slope m_θ depends on θ through the average size of partial quotients: large partial quotients drive faster denominator growth and hence a smaller m_θ. For $\theta = \phi$ (all partial quotients equal to 1), the asymptotic slope is $1/(2\log_2 \phi) \approx 0.72$; for $\theta = \log_2 3$ the empirical slope is approximately 0.45.

Conjecture 10.10 (Sub-universal bound)**.** *For each irrational $\theta > 1$ let m_θ^* denote the infimum of slopes m for which there exists b with $d \le m\log_2(e_d + o_d) + b$ for all $d \ge 0$. The supremum $m^* = \sup_\theta m_\theta^*$ satisfies $m^* < 2$; that is, the universal constants $(2, 1)$ of Theorem 10.9 are not tight.*

Conjecture 10.11 (Worst case at ϕ)**.** *The supremum m^* of Conjecture 10.10 is attained at $\theta = \phi$, with*

$$m_\phi^* \;=\; \frac{1}{2\log_2 \phi} \;\approx\; 0.72.$$

Equivalently, ϕ has the steepest asymptotic depth-vs-size curve among all irrationals $\theta > 1$.

DRAFT

Figure 2 shows that slope varies with partial-quotient size. Irrationals with small partial quotients (ϕ, $\sqrt{2}$) converge slowly, requiring more bits per depth step and producing steeper curves. Those with large partial quotients (π, $\log_3 8$) converge quickly and produce shallower curves. The universal bound $d \le 2\log_2(e_d + o_d) + 1$ of Theorem 10.9 applies to every curve. Figure 3 illustrates the per-target (m_θ, b_θ) pairs for $\log_2 3$ and ϕ, alongside the universal bound.

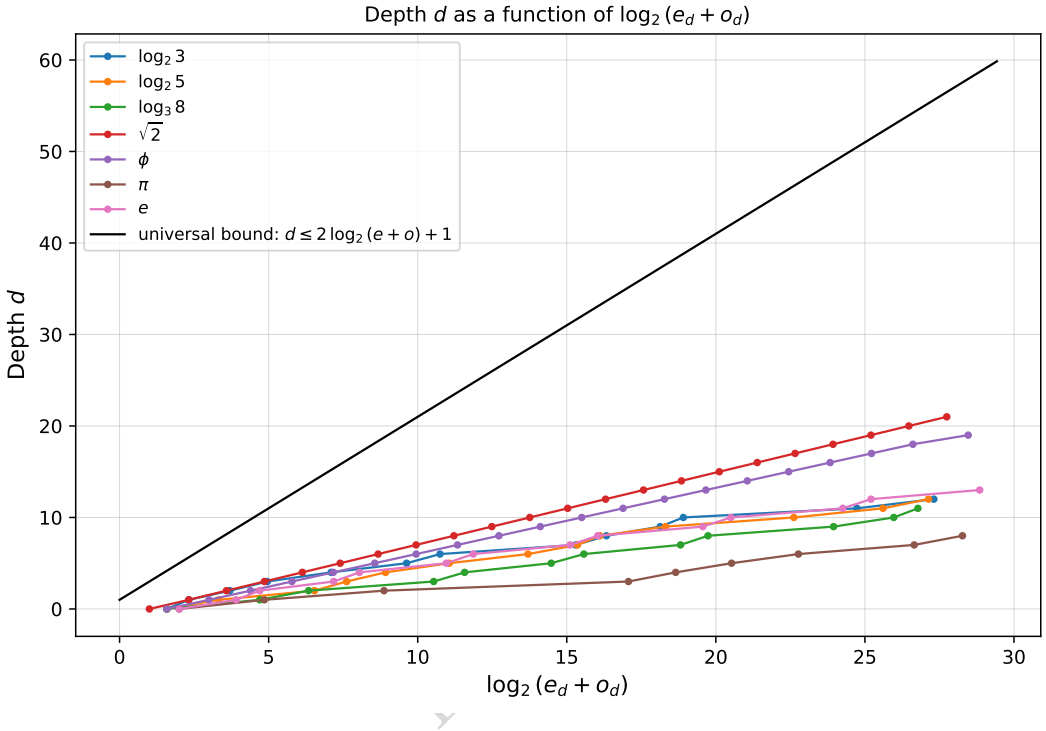


Figure 2: Depth d vs. $\log_2(e_d + o_d)$ for the better mediant-tree estimate at each depth, across seven irrationals. Each curve is approximately linear, consistent with Theorem 10.9. The solid black line is the universal proved bound $d \le 2\log_2(e_d + o_d) + 1$. The slope of each curve reflects partial-quotient size: irrationals with small partial quotients (e.g. ϕ, $\sqrt{2}$) converge slowly and have steeper curves; those with large partial quotients (e.g. π, $\log_3 8$) converge quickly and have shallower curves.

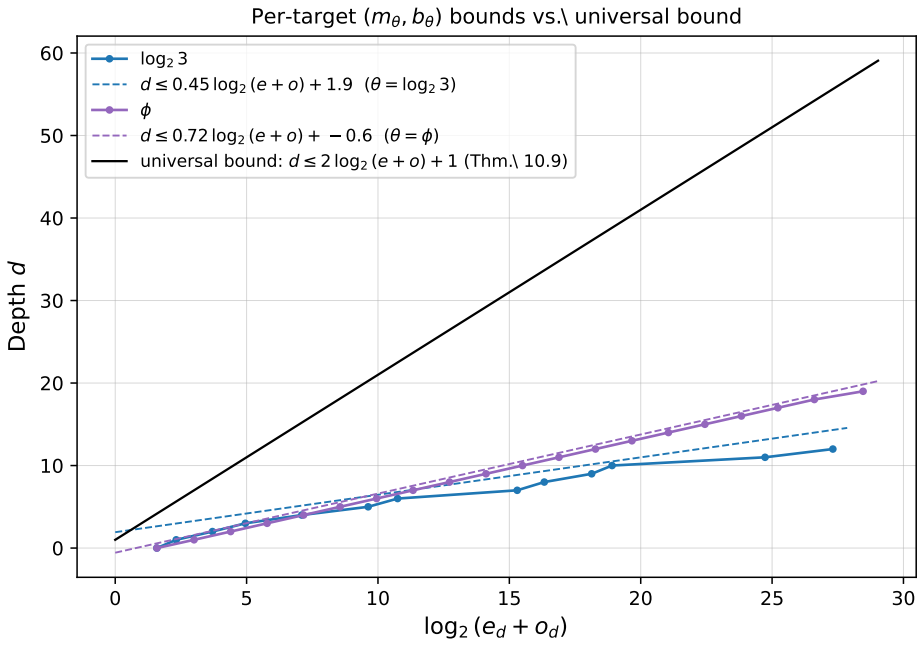


Figure 3: Per-target (m_θ, b_θ) bounds for $\theta = \log_2 3$ and $\theta = \phi$, together with the universal proved bound $d \le 2\log_2(e_d + o_d) + 1$ (Theorem 10.9, solid black line). Each dashed line is an empirical upper bound fitted to its own curve only; the slopes $m_{\log_2 3} \approx 0.45$ and $m_\phi \approx 0.72$ are both well below the universal ceiling $m = 2$.

DRAFT

11 Examples and Computational Results

Extended tables illustrating the algorithm on $\theta = \log_2 3$ and $\theta = \pi$ are given in Appendix B.

12 Relationship to Sturmian Sequences

A *Sturmian word* over $\{\mathsf{O}, \mathsf{E}\}$ with slope $\alpha \in (0, 1)$ is defined by

$$s_n = \lfloor (n+1)\alpha \rfloor - \lfloor n\alpha \rfloor, \qquad n \ge 0,$$

encoding the sequence as O when $s_n = 1$ and E when $s_n = 0$ (see also Section 4). Since $\alpha < 1$, we have $s_0 = \lfloor \alpha \rfloor = 0$, so the Sturmian word always begins with E.

12.1 Structural properties of mediant words

Before stating the main Sturmian identity we record four structural results about finite mediant words.

Theorem 12.1 (Every mediant word ends in O)**.** *For every estimate r produced by the mediant tree, the mediant word $\pi(r)$ ends in O.*

Proof. By structural induction on the mediant tree.

Base case. $\pi(e/1) = \mathsf{E}^e\mathsf{O}$, which ends in O.

Inductive step. Assume $\pi(r_0)$ and $\pi(r_1)$ both end in O. If $\Delta > 0$ then $\pi(r_{\text{new}}) = \pi(r_1)^w \cdot \pi(r_0)$, whose last symbol is the last symbol of $\pi(r_0)$, which is O by hypothesis. If $\Delta < 0$ then $\pi(r_{\text{new}}) = \pi(r_0) \cdot \pi(r_1)^w$, whose last symbol is the last symbol of the final copy of $\pi(r_1)$, again O. □

Theorem 12.2 (Overestimate criterion)**.** *Let $\theta > 1$ be irrational, let $r = (e, o)$ be a good estimate of θ (Definition 6.4), and set $L = e + o$. Let s be the Sturmian word with slope $\alpha = 1/(1 + \theta)$.*

(1) *The Sturmian symbol s_{L-1} satisfies*

$$s_{L-1} = \begin{cases} \mathsf{O} & \text{if } e > \theta o \text{ (} r \text{ an overestimate),} \\ \mathsf{E} & \text{if } e < \theta o \text{ (} r \text{ an underestimate).} \end{cases}$$

(2) *If moreover r is produced by the mediant tree, then $\pi(r)[-1] = \mathsf{O}$ (Theorem 12.1), and therefore $\pi(r)[-1] = s_{L-1}$ if and only if r is an overestimate.*

Proof. **Part (1).** The Sturmian word satisfies $s_j = \mathsf{O}$ iff the fractional part $f_j := \{(j+1)\alpha\}$ lies in $[0, \alpha)$. So $s_{L-1} = \mathsf{O}$ iff $\{(e + o)\alpha\} \in [0, \alpha)$.

Set $\Theta = o(e - \theta o)$, so $e = \theta o + \Theta/o$. Since $\alpha = 1/(1 + \theta)$:

$$(e + o)\alpha = o + \frac{\Theta\alpha}{o}.$$

DRAFT

Since o is a positive integer, $\{(e + o)\alpha\} = \{\Theta\alpha/o\}$. Since r is a good estimate, $|\Theta|/o < 1$; with $\alpha < 1/2$ we have $|\Theta\alpha/o| < \alpha < 1/2$, so:

- $\Theta > 0$: $\{\Theta\alpha/o\} = \Theta\alpha/o \in (0, \alpha)$, so $s_{L-1} = \mathsf{O}$.
- $\Theta < 0$: $\{\Theta\alpha/o\} = 1 + \Theta\alpha/o \in (1 - \alpha, 1) \subset (\alpha, 1)$, so $s_{L-1} = \mathsf{E}$.

Part (2). For a tree estimate, $\pi(r)[-1] = \mathsf{O}$ by Theorem 12.1. Combined with Part (1), $\pi(r)[-1] = s_{L-1}$ iff both equal O, i.e. iff r is an overestimate. □

Remark 12.3. When $r = \text{Mediant}(r_{2d}, r_{2d+1}, w^+)$ is the child at the bracketing weight, the overestimate condition $e > \theta o$ is equivalent to $\Delta(r_{2d}, r_{2d+1}) > 0$: since w^+ is the bracketing weight, $\text{sgn}(e - \theta o) = \text{sgn}(f(w^+)) = \text{sgn}(B) = \text{sgn}(\Delta)$ (where $f(t) = A + tB$ as in the proof of Theorem 8.1, and $\Delta = \Delta(r_{2d}, r_{2d+1})$ is the determinant of the *parent pair*). For the companion child $r = \text{Mediant}(r_{2d}, r_{2d+1}, w^-)$ (when $w^+ = 1$, the companion is r_{2d} itself), the relationship is reversed. The sign of Δ of the parent pair is therefore not a uniform criterion for a general good estimate; the sign of $\Theta = o(e - \theta o)$ is.

Theorem 12.4 (Universal EE prefix)**.** *For every non-leaf estimate r produced by the mediant tree (i.e. $|\pi(r)| > \lceil\theta\rceil + 1$), the mediant word begins $\pi(r)[0{:}2] = \mathsf{EE}$.*

Proof. By Lemma 9.3, $\pi(r) = \mathrm{Ch}^-(e, o)$. Every non-leaf estimate $r = e/o$ lies strictly between the two leaf estimates $\lfloor\theta\rfloor/1$ and $\lceil\theta\rceil/1$, so in particular $e/o > \lfloor\theta\rfloor \geq 1$, giving $e > o$. By Definition 5.1, $\mathrm{Ch}^-(e, o)[i] = \mathsf{O}$ iff $\lfloor(i+1)o/(e+o)\rfloor > \lfloor i \cdot o/(e+o)\rfloor$. For $i = 0$: $1 \cdot o/(e+o) < 1$ (since $o < e+o$), so $\lfloor o/(e+o)\rfloor = 0 = \lfloor 0\rfloor$, giving $\pi(r)[0] = \mathsf{E}$. For $i = 1$: $e > o$ implies $e + o > 2o$, so $2o/(e+o) < 1$, giving $\lfloor 2o/(e+o)\rfloor = 0$, and $\pi(r)[1] = \mathsf{E}$. Hence $\pi(r)[0{:}2] = \mathsf{EE}$. □

Theorem 12.5 (Palindrome invariant)**.** *For every estimate r produced by the mediant tree with $L = |\pi(r)| \geq 2$, the interior slice $\pi(r)[1{:}L{-}1]$ is a palindrome.*

Proof. By Lemma 9.3, $\pi(r) = \mathrm{Ch}^-(e, o)$ with $L = e{+}o$. By Definition 5.1, $\mathrm{Ch}^-(e, o) = \mathsf{E}{\cdot}U{\cdot}\mathsf{O}$ where U is the palindromic interior (Berstel [5], Chapter 1). Therefore $\pi(r)[1{:}L{-}1] = U$ is a palindrome. □

Theorem 12.6 (Sturmian identity)**.** *Let $\theta > 1$ be irrational, $\alpha = 1/(\theta+1)$, and let s be the Sturmian sequence with slope α. For every estimate r produced by the mediant tree, with mediant word $\pi(r)$ of length N,*

$$\pi(r)[i] = s_i \qquad \text{for all } 0 \leq i \leq N-2.$$

The terminal position $i = N-1$ is excluded: by Theorem 12.1 the mediant always emits O there, while by Theorem 12.2 the Sturmian emits O if r is an overestimate and E if r is an underestimate.

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Proof. By Lemma 9.3, $\pi(r) = \mathrm{Ch}^-(e, o)$ with $L = e + o$. By Definition 5.1, $\mathrm{Ch}^-(e, o)[i] = \mathsf{O}$ iff $\lfloor(i+1)o/L\rfloor > \lfloor i \cdot o/L\rfloor$, and the Sturmian word satisfies $s_i = \mathsf{O}$ iff $\lfloor(i+1)\alpha\rfloor > \lfloor i\alpha\rfloor$. It suffices to show $\lfloor n \cdot o/L\rfloor = \lfloor n\alpha\rfloor$ for all $1 \leq n \leq L-1$.

The estimate $r = e/o$ belongs to a bounded pair (r_{2d}, r_{2d+1}) at some depth d, with the two fractions $\alpha_0 = o_0/L_0$ and $\alpha_1 = o_1/L_1$ (where $L_k = e_k + o_k$) satisfying $\alpha_0 < \alpha < \alpha_1$ and the Farey adjacency condition $o_1L_0 - o_0L_1 = \pm 1$. In particular o/L is one of α_0 or α_1, and $L = L_k$ for the appropriate k.

Key lemma (Farey gap). For $1 \leq n \leq L-1$, no fraction with denominator $< L$ lies strictly between α and o/L. *Proof:* Suppose m/n with $0 < n < L$ lies strictly between α and o/L. Then m/n lies strictly between α_0 and α_1. By the Farey mediant property, any fraction strictly between two Farey-adjacent fractions $\alpha_0 = o_0/L_0$ and $\alpha_1 = o_1/L_1$ (with $|o_1L_0 - o_0L_1| = 1$) has denominator at least $L_0 + L_1 > L$, contradicting $n < L$.

Agreement. Fix $1 \leq n \leq L-1$ and suppose for contradiction that $\lfloor n\alpha\rfloor \neq \lfloor n \cdot o/L\rfloor$. Then there exists an integer m with $n\alpha < m \leq n \cdot o/L$ or $n \cdot o/L < m \leq n\alpha$ (the two cases are symmetric). In either case m/n has denominator $n < L$. Note that $m/n = o/L$ is impossible since $\gcd(o, L) = 1$ and $0 < n < L$, and $m/n = \alpha$ is impossible since α is irrational. Hence m/n lies *strictly* between α and o/L, contradicting the Key lemma. Hence $\lfloor n\alpha\rfloor = \lfloor n \cdot o/L\rfloor$ for all $1 \leq n \leq L-1$, and therefore $\mathrm{Ch}^-(e, o)[i] = s_i$ for $0 \leq i \leq L-2$.

Terminal position. The exclusion of $i = L-1$ follows from Theorems 12.1 and 12.2: $\pi(r)$ ends in O, while $s_{L-1} = \mathsf{O}$ iff r is an overestimate ($e/o > \theta$). □

Corollary 12.7 (Mismatch characterisation)**.** *The mediant word $\pi(r)$ and the Sturmian prefix $s[0{:}L]$ are identical if and only if r is an overestimate ($e/o > \theta$). When r is an underestimate ($e/o < \theta$), they agree at every position except $i = L-1$, where the mediant emits* O *and the Sturmian emits* E*.*

The single terminal discrepancy is an artefact of truncation, not a genuine disagreement: the mediant sequence closes its last primitive word with O at position $N-1$, whereas the Sturmian word — which continues indefinitely — may still be in the interior of a longer word at that depth. As the depth increases and $N \to \infty$, both sequences converge to the same infinite word.

Remark 12.8. The identification of finite approximants of a Sturmian word with lower Christoffel words is classical (Lothaire [3], Berstel et al. [5]). The novel content of Theorem 12.6 is the proof via the mediant tree, which gives an explicit construction of these approximants together with a precise characterisation of the terminal discrepancy (the last-letter criterion, Theorems 12.1 and 12.2).

This has been verified computationally for several irrational values of θ, including $\sqrt{2}$, $\log_2 3$, $\log_2 5$, ϕ, and π. For $\theta = \log_2 3$ the verification extends to sequences of length up to 27,869,189 symbols: at every depth the two sequences agree everywhere except possibly the last symbol, where the mediant emits O and the Sturmian emits E.

DRAFT

Remark 12.9 (The mediant sequence is the better finite approximation). At depths where the two sequences differ, the mediant's finite mediant word is a closer approximation to $\alpha = 1/(\theta+1)$ in terms of running O-density. The terminal O emitted by the mediant sequence gives a density slightly *above* α, whereas the Sturmian's terminal E gives a density substantially *below* α.

This reflects a structural asymmetry in the two methods. The Sturmian algorithm at position $N-1$ is mid-word in a longer primitive word, so it emits E — inherently favouring the above-θ estimate and leaving the running density below α. The mediant tree, by contrast, explicitly selects at each step the estimate with smaller $|\varepsilon|$, i.e. r_{2d+1}, whichever side of θ it falls on. The terminal O is a consequence of the Christoffel word structure: every lower Christoffel word (and hence every mediant word; Theorem 12.1) ends in O, regardless of which side of θ the estimate falls on.

For $\theta = \log_2 3$, the improvement at the discrepancy depths is substantial: at $N = 1719$ the mediant density error is 1.4×10^{-8} against the Sturmian's 5.8×10^{-4} — roughly four orders of magnitude better.

The O-density of $\pi(r)$ for an estimate e/o is

$$\frac{o}{e+o} = \frac{1}{e/o+1} \xrightarrow{e/o \to \theta} \frac{1}{\theta+1} = \alpha,$$

consistent with the Sturmian slope.

Efficiency and lazy generation. Although the mediant words are Sturmian, the mediant tree is a far more efficient way to generate them. To produce a prefix of N symbols, the naive Sturmian algorithm must evaluate $s_n = \lfloor (n+1)\alpha \rfloor - \lfloor n\alpha \rfloor$ at each of the N positions, requiring $O(N)$ real-arithmetic operations.

The mediant algorithm instead builds a compressed representation of the same prefix. Each iteration requires a constant number of real-arithmetic operations to compute the next bounded pair and weight; after k iterations the tree encodes a Sturmian prefix of length $N = e_{2d+1} + o_{2d+1}$. N grows exponentially with k (Theorem 10.9), so $O(\log N)$ iterations suffice to encode a prefix of length N, each requiring $O(1)$ oracle evaluations.

Once the tree is built, enumerating the N symbols requires *no real arithmetic at all*: each node specifies an integer weight w and two child words, and reading off the sequence is a recursive depth-first traversal governed entirely by integer counts. No auxiliary memory beyond the call stack is required.

OO-avoidance. Since $\theta > 1$ implies $\alpha = 1/(\theta+1) < 1/2$, the Sturmian word avoids OO: two consecutive O's would require $\lfloor (n+2)\alpha \rfloor - \lfloor n\alpha \rfloor = 2$, but $2\alpha < 1$ makes this impossible. The mediant tree gives an independent structural proof: every mediant word ends in O and begins with E (since $e \geq 1$; Lemma 9.3 and Definition 5.1), so every concatenation boundary within a branch word $v^w \cdot u$ or $u \cdot v^w$ reads $\cdots$OE$\cdots$; OO never arises at any boundary, and by induction never arises anywhere.

Remark 12.10. The propagated weights w_d^+ are related to the partial quotients of the continued fraction expansion of θ, though they are not in general identical to them. A large weight at depth d corresponds to a large partial quotient, producing a long run of one symbol type in the mediant word — the same phenomenon that gives rise to the irregular spacing of O's and E's in an irrational Sturmian word. Conversely, small weights correspond to small partial quotients, and the mediant word advances by only a short block at each such step.

DRAFT

13 Extension to $\theta \leq 1$

The algorithm of Sections 7–9 requires $\theta > 1$. We now show how to handle all other real θ by algebraic reduction.

If one were content to accept OO in the mediant word, no special treatment would be needed: the algorithm and the Sturmian definition extend formally to any $\theta > 0$, and for $\theta < 1$ the slope $\alpha = 1/(1+\theta) > 1/2$ makes OO unavoidable. The reductions below are motivated by applications (described in [7]) that require OO-free mediant words.

Notation. Let $r(\theta)$ denote the rational approximant produced by the algorithm for $\theta > 1$, i.e. the sequence of bounded pairs (r_{2d}, r_{2d+1}) and their limit. Each subsection below defines $r(\theta)$ for the remaining cases by reducing to a $\theta' > 1$ and applying a simple transformation.

13.1 Case $0 < \theta < \frac{1}{2}$

For $\theta \in (0, \frac{1}{2})$, define $\theta' = (1-\theta)/\theta > 1$ and set

$$r(\theta) = \frac{o}{e+o},$$

where $e/o = r(\theta')$. That is, if $r(\theta') = e/o$ then $r(\theta)$ is obtained by swapping the role of e and o: numerator o (O-count), denominator $e+o$ (total length).

Mediant word. The mediant word for θ' has O-density

$$\frac{1}{1+\theta'} = \frac{1}{1+(1-\theta)/\theta} = \theta,$$

which is less than $\frac{1}{2}$ since $\theta < \frac{1}{2}$. Consequently the OO-avoidance result of Section 12 applies: the mediant word for θ' is OO-free, and its O-density converges to θ.

Sturmian identity. The Sturmian word with slope $\alpha' = \theta$ is the infinite mediant word produced by the mediant tree for θ' (Theorem 12.6). This Sturmian word encodes θ directly: the O-density of any long prefix approximates θ with the same precision as the underlying mediant approximation of θ'.

DRAFT

13.2 Case $\frac{1}{2} < \theta < 1$

For $\theta \in (\frac{1}{2}, 1)$, define $\theta' = \theta/(1-\theta) > 1$ and set

$$r(\theta) = \frac{e}{e+o},$$

where $e/o = r(\theta')$.

Mediant word. The mediant word for θ' has O-density

$$\frac{1}{1+\theta'} = \frac{1}{1+\theta/(1-\theta)} = 1-\theta < \tfrac{1}{2},$$

so it is OO-free and its Sturmian slope is $1-\theta$. The sequence encodes θ'; the rational approximant $e/(e+o)$ to θ is read off from the E- and O-counts of that sequence.

13.3 Case $\theta < 0$

For $\theta < 0$, set

$$r(\theta) = -r(-\theta).$$

13.4 Degenerate cases $\theta \in \{0, \frac{1}{2}, 1\}$

These three values sit at the boundary between regimes and are defined directly:

$$r(0) = 0/1, \qquad r(\tfrac{1}{2}) = 1/2, \qquad r(1) = 1/1.$$

The corresponding mediant words are E^∞, $(\mathsf{EO})^\infty$, and $(\mathsf{EO})^\infty$ respectively (at $\theta = 1$ the O-density is $1/2$, consistent with the $\theta \to 1^+$ limit of $1/(1+\theta) \to 1/2$). Note the one-sided limit: as $\theta \to 1^-$ in the $(\frac{1}{2}, 1)$ regime, the Sturmian slope is $1 - \theta \to 0$, so the words tend to E^∞, not $(\mathsf{EO})^\infty$.

Summary. Table 1 collects the reductions for all ranges of θ.

Table 1: Reductions to the $\theta > 1$ algorithm for all ranges of θ

Range of θ	Reduction θ'	Approximant to θ	O-density
$\theta > 1$	$\theta' = \theta$	e/o	$1/(1+\theta) < \frac{1}{2}$
$0 < \theta < \frac{1}{2}$	$\theta' = (1-\theta)/\theta$	$o/(e+o)$	$\theta < \frac{1}{2}$
$\frac{1}{2} < \theta < 1$	$\theta' = \theta/(1-\theta)$	$e/(e+o)$	$1 - \theta < \frac{1}{2}$
$\theta < 0$	$\theta' = -\theta$ (or above)	negate	—
$\theta = 0$	trivial: word E^∞	$0/1$	0
$\theta = \frac{1}{2}$	trivial: word $(\mathsf{EO})^\infty$	$1/2$	$\frac{1}{2}$
$\theta = 1$	trivial: word $(\mathsf{EO})^\infty$	$1/1$	$\frac{1}{2}$

DRAFT

Figures 4 and 5 plot the reduction map and the resulting Sturmian slope across all regimes.

14 Conclusion and Open Problems

We have presented a theory of rational approximation for real numbers based on bounded pairs and the weighted mediant construction. The core algorithm applies to $\theta > 1$; all other real numbers are handled by algebraic reduction (Section 13).

What is new. The weighted mediant step is the classical Stern–Brocot step; the novelty is Theorem 8.1, which identifies the *unique optimal weight* w^+ via a sign-change bracket on the linear error function, giving a closed-form formula; that each iterate lies on the Stern–Brocot path of θ is a consequence of the weighted-mediant Farey structure (Theorem 10.3). Lemma 9.3 (every mediant word is a lower Christoffel word) follows from the standard Christoffel concatenation theorem (Berstel et al. [5]); the contribution is the identification that the mediant algorithm *naturally* produces lower Christoffel words, without any explicit reference to the Christoffel tree. The Sturmian identity (Theorem 12.6) and the palindrome invariant (Theorem 12.5) are consequences of the Christoffel word theory (Lothaire [3], Berstel et al. [5]); the contribution is a direct proof via the mediant construction, together with the last-letter criterion (Theorems 12.1 and 12.2) which is new. The mediant tree taken as

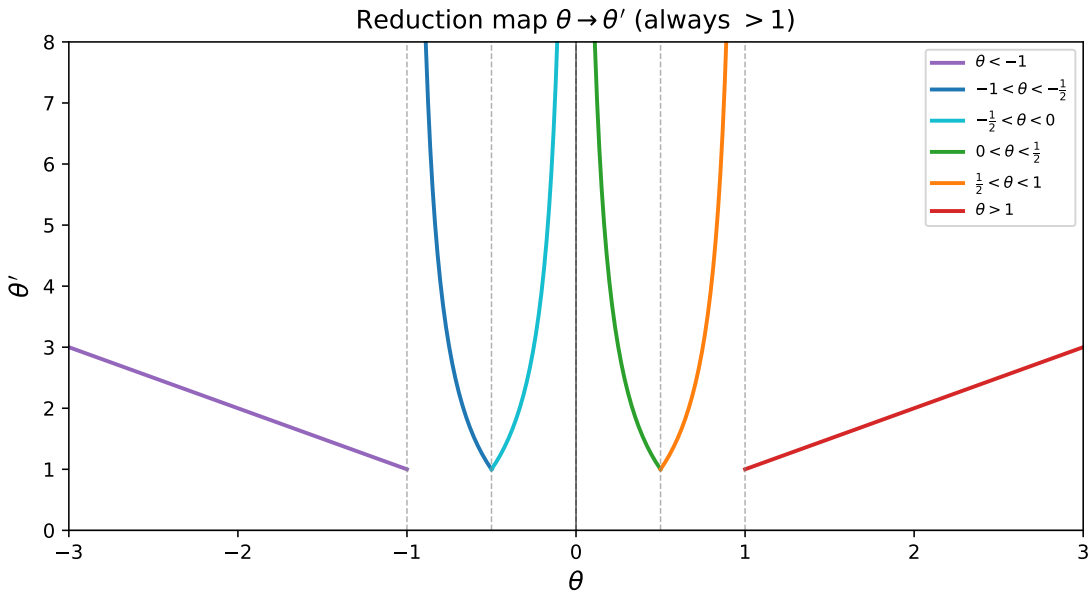


Figure 4: The reduction map $\theta \to \theta'$, which is always greater than 1. Colours identify the regimes; vertical dashed lines mark the boundary values $\theta \in \{-1, -\frac{1}{2}, 0, \frac{1}{2}, 1\}$.

DRAFT

a whole is a *compressed transducer*: it encodes the construction history of a length-N Sturmian prefix in $O(\log N)$ iterations (Conjecture 10.9; Section 10), each requiring $O(1)$ oracle evaluations, after which the symbols are enumerated by a purely integer recursive traversal.

Open problems.

1. **Characterisation of the MT fractions.** Every MT fraction is a convergent or semi-convergent of θ; the MT visits the subsequence of those whose consecutive mediants straddle θ. Characterise this subsequence in terms of the continued-fraction expansion of θ, and determine when an MT fraction equals a CF convergent.

2. **Cost comparison with CF.** The CF algorithm produces convergents in $O(\log N)$ arithmetic steps; the MT does likewise but visits different fractions. Give a precise comparison of the approximation quality per step between the two algorithms.

3. **Linear depth bound.** See Conjecture 10.9 in Section 10.

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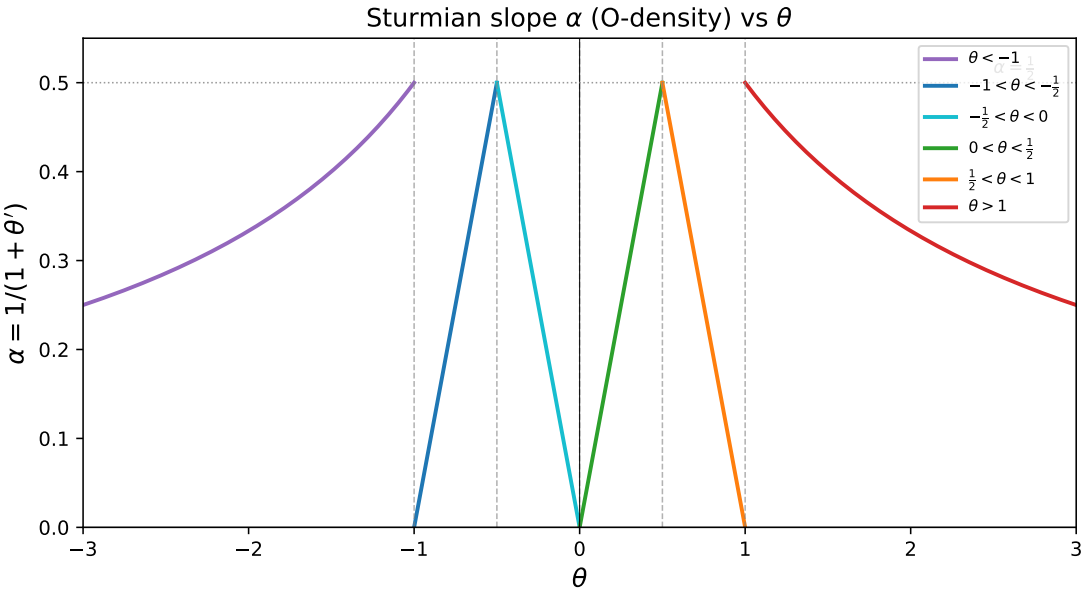


Figure 5: The Sturmian slope $\alpha = 1/(1+\theta')$ (the O-density of the mediant word), which is always at most $\frac{1}{2}$ (equalling $\frac{1}{2}$ at $\theta = 1$). Colours and boundary lines as in Figure 4.

DRAFT

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[6] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, fifth edition, Oxford University Press, 1979. (See §11.10: Hurwitz's theorem and the role of ϕ as the hardest irrational to approximate by rationals.)

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A Proof of Theorem 8.1

We prove the closed-form formula $w^+ = \lfloor -A/B \rfloor + 1$ directly from the bracketing condition, using only a linear sign argument.

We have a bounded pair (r_{2d}, r_{2d+1}) bracketing θ, with

$$A = e_{2d} - \theta\, o_{2d}, \qquad B = e_{2d+1} - \theta\, o_{2d+1}.$$

Since r_{2d} and r_{2d+1} are on opposite sides of θ we have $\operatorname{sgn}(A) = -\operatorname{sgn}(B)$, so $A/B < 0$ and $c := -A/B > 0$. The two weighted mediants are

$$r_- = \frac{e_{2d} + (w-1)\, e_{2d+1}}{o_{2d} + (w-1)\, o_{2d+1}}, \qquad r_+ = \frac{e_{2d} + w\, e_{2d+1}}{o_{2d} + w\, o_{2d+1}},$$

with signed errors (numerators of ε^-, ε^+ up to positive denominators)

$$A + (w-1)B \qquad \text{and} \qquad A + wB.$$

Bracketing condition. The pair (r_-, r_+) brackets θ iff ε^- and ε^+ have opposite signs, i.e. $(A+(w-1)B)(A+wB) < 0$. Writing $c = -A/B$, this product equals $B^2\,(w-1-c)(w-c)$, which is negative iff $w-1-c$ and $w-c$ have opposite signs, i.e.

$$w - 1 < c < w,$$

and since $c \notin \mathbb{Z}$ (as θ is irrational, $A + nB \neq 0$ for all integers n) this is equivalent to $w - 1 = \lfloor c \rfloor$, i.e. $w = \lfloor c \rfloor + 1$. Since c is never an integer, exactly one integer w satisfies the condition, namely

$$w^+ = \lfloor -A/B \rfloor + 1.$$

We also write $w^- = w^+ - 1 = \lfloor -A/B \rfloor$ for the companion weight.

Expression in terms of Δ and Θ. Recall $\Theta = o_{2d+1}B$ and $A\, o_{2d+1} - B\, o_{2d} = -\Delta$, giving $A = (Bo_{2d} - \Delta)/o_{2d+1}$, so

$$\frac{-A}{B} = \frac{\Delta}{o_{2d+1}B} - \frac{o_{2d}}{o_{2d+1}} = \frac{\Delta}{\Theta} - \frac{o_{2d}}{o_{2d+1}},$$

and therefore

$$w^+ = \left\lfloor \frac{\Delta}{\Theta} - \frac{o_{2d}}{o_{2d+1}} \right\rfloor + 1,$$

completing the proof of Theorem 8.1.

Termination. When $\Theta = 0$, r_{2d+1} is an exact rational representation of θ; no further improvement is possible and the algorithm terminates.

B Examples and Computational Results

B.1 Worked example: mediant words for $\log_2 3$

Example B.1 (Mediant words and trees for $\log_2 3$)**.** Starting from the initial bounded pair $(r_0, r_1) = (1/1,\ 2/1)$ with $\theta = \log_2 3 \approx 1.58496$, the algorithm produces the following sequence of bounded pairs. At each depth d, propagated weight w_d^+ is used to compute the two candidates $r_- = \text{Mediant}(r_{2d}, r_{2d+1}, w_d^+ - 1)$ and $r_+ = \text{Mediant}(r_{2d}, r_{2d+1}, w_d^+)$; the better estimate (smaller $|\varepsilon|$) becomes r_{2d+3}.

DRAFT

d	r_{2d}	r_{2d+1}	$\varepsilon_{r_{2d}}$	$\varepsilon_{r_{2d+1}}$	w_d^+	r_-	r_+
0	1/1	2/1	-0.369	$+0.262$	2	3/2	5/3
1	3/2	5/3	-0.054	$+0.052$	1	3/2	8/5
2	3/2	8/5	-0.054	$+0.009$	3	19/12	27/17
3	27/17	19/12	$+0.002$	-0.001	3	65/41	84/53

The leaf words are $\pi(r_0) = \texttt{EO}$ and $\pi(r_1) = \texttt{EEO}$. Reading the word decomposition bottom-up:

Level 1 ($d = 0$, $w_0^+ = 2$, $\Delta > 0$, $\pi(r_3) = \pi(r_1)^2 \cdot \pi(r_0)$, $\pi(r_2) = \pi(r_1)^1 \cdot \pi(r_0)$):

$$\underbrace{\underbrace{\texttt{EEO}}_{\pi(r_1)}\,\underbrace{\texttt{EEO}}_{\pi(r_1)}\,\underbrace{\texttt{EO}}_{\pi(r_0)}}_{\pi(r_3)=\texttt{EEOEEOEO}} \qquad \underbrace{\underbrace{\texttt{EEO}}_{\pi(r_1)}\,\underbrace{\texttt{EO}}_{\pi(r_0)}}_{\pi(r_2)=\texttt{EEOEO}}$$

Level 2 ($d = 1$, $w_1^+ = 1$, $\Delta > 0$, $\pi(r_5) = \pi(r_3)^1 \cdot \pi(r_2)$, $\pi(r_4) = \pi(r_2)$):

$$\underbrace{\underbrace{\texttt{EEOEEOEO}}_{\pi(r_3)}\,\underbrace{\texttt{EEOEO}}_{\pi(r_2)}}_{\pi(r_5)=\texttt{EEOEEOEOEEOEO}} \qquad \underbrace{\underbrace{\texttt{EEOEO}}_{\pi(r_2)}}_{\pi(r_4)=\texttt{EEOEO}}$$

Level 3 ($d = 2$, $w_2^+ = 3$, $\Delta > 0$, $\pi(r_7) = \pi(r_5)^2 \cdot \pi(r_4)$):

$$\underbrace{\underbrace{\texttt{EEOEEOEOEEOEO}}_{\pi(r_5)}\,\underbrace{\texttt{EEOEEOEOEEOEO}}_{\pi(r_5)}\,\underbrace{\texttt{EEOEO}}_{\pi(r_4)}}_{\pi(r_7)}$$

DRAFT

The same estimates as mediant trees (depths $d = 1$ and $d = 2$):

$$\begin{aligned}
r_2 &= 3/2 = \text{Mediant}(\underbrace{1}_{\text{leaf}},\ \underbrace{2}_{\text{leaf}},\ w = 1),\\
r_3 &= 5/3 = \text{Mediant}(\underbrace{1}_{\text{leaf}},\ \underbrace{2}_{\text{leaf}},\ w = 2),\\
r_4 &= 3/2 = \text{Mediant}(\text{Mediant}(1,\,2,\,1),\ \text{Mediant}(1,\,2,\,2),\ w = 0),\\
r_5 &= 8/5 = \text{Mediant}(\text{Mediant}(1,\,2,\,1),\ \text{Mediant}(1,\,2,\,2),\ w = 1).
\end{aligned}$$

A weight of $w = 0$ means the left subtree is used directly (contributing zero copies of the right subtree).

B.2 Approximating $\log_2 3$

Table 2 shows 22 steps of the algorithm for $\theta = \log_2 3$, starting from $r_0 = 1/1$, $r_1 = 2/1$. Note the large weight $w = 24$ at depth $d = 6$ and $w = 55$ at depth $d = 10$, corresponding to large partial quotients of the continued fraction of $\log_2 3$, which produce large jumps in precision.

Table 2: Successive bounded pairs approximating $\log_2 3 \approx 1.58496\ldots$

d	r_{2d}	r_{2d+1}	$\varepsilon_{r_{2d}}$	$\varepsilon_{r_{2d+1}}$	w^+	r_-	r_+
0	1/1	2/1	-0.369	$+0.262$	2	3/2	5/3
1	3/2	5/3	-0.054	$+0.052$	1	3/2	8/5
2	3/2	8/5	-0.054	$+0.009$	3	19/12	27/17
3	27/17	19/12	$+0.002$	-0.001	3	65/41	84/53
4	65/41	84/53	$+2.54\times10^{-4}$	-3.59×10^{-5}	6	485/306	569/359
5	485/306	569/359	$+3.04\times10^{-6}$	-2.70×10^{-6}	1	485/306	1054/665
6	485/306	1054/665	$+3.04\times10^{-6}$	-5.98×10^{-8}	24	24727/15601	25781/16266
7	25781/16266	24727/15601	-1.42×10^{-9}	$+1.06\times10^{-9}$	2	50508/31867	75235/47468
8	75235/47468	50508/31867	$+2.10\times10^{-10}$	-2.08×10^{-10}	2	125743/79335	176251/111202
9	125743/79335	176251/111202	$+4.20\times10^{-11}$	-2.95×10^{-11}	2	301994/190537	478245/301739
10	478245/301739	301994/190537	-1.07×10^{-11}	$+3.08\times10^{-13}$	55	16785921/10590737	17087915/10781274
11	16785921/10590737	17087915/10781274	-4.50×10^{-15}	$+1.03\times10^{-15}$	5	85137581/53715833	102225496/64497107
12	102225496/64497107	85137581/53715833	$+1.23\times10^{-16}$	-5.88×10^{-17}	3	272500658/171928773	357638239/225644606
13	272500658/171928773	357638239/225644606	$+9.47\times10^{-18}$	-6.79×10^{-18}	2	630138897/397573379	987777136/623217985
14	987777136/623217985	630138897/397573379	-2.30×10^{-18}	$+2.42\times10^{-19}$	15	9809721694/6189245291	10439860591/6586818670
15	9809721694/6189245291	10439860591/6586818670	-1.41×10^{-20}	$+1.40\times10^{-21}$	10	103768467013/65470613321	114208327604/72057431991
16	114208327604/72057431991	103768467013/65470613321	$+6.96\times10^{-23}$	-6.41×10^{-23}	2	217976794617/137528045312	321745261630/202998658633
17	321745261630/202998658633	217976794617/137528045312	-1.67×10^{-23}	$+5.95\times10^{-24}$	5	1193652440098/753110839881	1411629234715/890638885193
18	1411629234715/890638885193	1193652440098/753110839881	$+7.97\times10^{-25}$	-1.44×10^{-25}	7	8573543875303/5409303924479	9767196315401/6162414764360
19	8573543875303/5409303924479	9767196315401/6162414764360	$+1.11\times10^{-26}$	-7.83×10^{-27}	2	18340740190704/11571718688839	28107936506105/17734133453199
20	28107936506105/17734133453199	18340740190704/11571718688839	-2.06×10^{-27}	$+1.01\times10^{-27}$	4	83130157078217/52449289519716	101470897268921/64021008208555
21	101470897268921/64021008208555	83130157078217/52449289519716	$+1.63\times10^{-28}$	-2.51×10^{-29}	8	683381996816440/431166034846567	766512153894657/483615324366283

B.3 Approximating π

Table 3 shows 22 steps for $\theta = \pi$, starting from $r_0 = 4/1$, $r_1 = 3/1$. The classical approximant $355/113$ is produced from the depth-1 pair and appears as r_5, the better member of the depth-2 pair; it has $\varepsilon \approx +8.49\times10^{-8}$, and the weight $w = 293$ at depth $d = 2$ produces an immediate jump of roughly three orders of magnitude (from $|\varepsilon| \approx 8.49\times10^{-8}$ to $|\varepsilon| \approx 1.06\times10^{-10}$).

Table 3: Successive bounded pairs approximating $\pi \approx 3.14159\ldots$

d	r_{2d}	r_{2d+1}	$\varepsilon_{r_{2d}}$	$\varepsilon_{r_{2d+1}}$	w^+	r_-	r_+
0	4/1	3/1	$+0.273$	-0.045	7	22/7	25/8
1	25/8	22/7	-5.28×10^{-3}	$+4.02\times10^{-4}$	15	333/106	355/113
2	333/106	355/113	-2.65×10^{-5}	$+8.49\times10^{-8}$	293	103993/33102	104348/33215
3	103993/33102	104348/33215	-1.84×10^{-10}	$+1.06\times10^{-10}$	2	208341/66317	312689/99532
4	208341/66317	312689/99532	-3.89×10^{-11}	$+9.28\times10^{-12}$	3	833719/265381	1146408/364913
5	833719/265381	1146408/364913	-2.77×10^{-12}	$+5.13\times10^{-13}$	4	4272943/1360120	5419351/1725033
6	4272943/1360120	5419351/1725033	-1.29×10^{-13}	$+7.05\times10^{-15}$	15	80143857/25510582	85563208/27235615
7	85563208/27235615	80143857/25510582	$+2.74\times10^{-16}$	-1.84×10^{-16}	2	165707065/52746197	245850922/78256779
8	165707065/52746197	245850922/78256779	$+5.22\times10^{-17}$	-2.49×10^{-17}	2	411557987/131002976	657408909/209259755
9	411557987/131002976	657408909/209259755	$+6.16\times10^{-18}$	-5.45×10^{-18}	1	411557987/131002976	1068966896/340262731
10	411557987/131002976	1068966896/340262731	$+6.16\times10^{-18}$	-9.77×10^{-19}	3	2549491779/811528438	3618458675/1151791169
11	2549491779/811528438	3618458675/1151791169	$+1.76\times10^{-19}$	-1.65×10^{-19}	1	2549491779/811528438	6167950454/1963319607
12	2549491779/811528438	6167950454/1963319607	$+1.76\times10^{-19}$	-2.43×10^{-20}	3	14885392687/4738167652	21053343141/6701487259
13	14885392687/4738167652	21053343141/6701487259	$+9.94\times10^{-21}$	-8.33×10^{-23}	85	17833366216531/567663097408	1804419559672/574364584667
14	1804419559672/574364584667	17833366216531/567663097408	-5.85×10^{-25}	$+3.91\times10^{-25}$	2	3587785776203/1142027682075	5371151992734/1709690779483
15	3587785776203/1142027682075	5371151992734/1709690779483	-1.00×10^{-25}	$+6.28\times10^{-26}$	2	8958937768937/2851718461558	14330089761671/4561409241041
16	14330089761671/4561409241041	8958937768937/2851718461558	$+2.20\times10^{-26}$	-2.46×10^{-27}	15	139755218526789/44485467702853	148714156295726/47337186164411
17	148714156295726/47337186164411	139755218526789/44485467702853	-9.99×10^{-29}	$+5.13\times10^{-29}$	3	428224593349304/136308121570117	567979811876093/180793589272970
18	567979811876093/180793589272970	428224593349304/136308121570117	$+1.17\times10^{-29}$	-1.21×10^{-30}	13	5706674932067741/1816491048114374	6134899525417045/1952799169684491
19	5706674932067741/1816491048114374	6134899525417045/1952799169684491	$+7.43\times10^{-32}$	-1.55×10^{-32}	5	30246273033735921/9627687726852338	36381172559152966/11580486896536829
20	30246273033735921/9627687726852338	36381172559152966/11580486896536829	$+1.45\times10^{-33}$	-1.40×10^{-33}	1	30246273033735921/9627687726852338	66627445592888887/21208174623389167
21	30246273033735921/9627687726852338	66627445592888887/21208174623389167	$+1.45\times10^{-33}$	-1.07×10^{-34}	7	430010946591069243/136876735467187340	4966383921839581 30/158084910090576507

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Disclosures

Use of Artificial Intelligence

Large language models played a substantial role in the development of this paper, and the author considers it important to describe that role honestly.

Review and gap identification. Claude (Anthropic) and Gemini (Google) were used as critical reviewers across multiple structured review rounds. Each model was presented with the manuscript and asked to identify mathematical gaps, proof obligations, notational

inconsistencies, and expository weaknesses. The resulting issue lists directly drove successive revisions and are documented in the paper's revision history.

Theorem formulation. The boundary between the author identifying a result and an AI model identifying it is not always sharp. In several instances the need for a theorem or lemma was recognised during AI-assisted review rather than by the author independently. The author regards this as a genuine intellectual contribution by the models involved, not merely automated checking.

Proof development. On several occasions, Aristotle (the code name used for the AI reviewer in this project's review workflow) did not merely flag a conjecture as unproven but supplied the proof that elevated it to a theorem. Theorem 10.5 and Theorem 10.7 are examples where the final rigorous proofs emerged from this collaborative process. Claude also drafted proofs and theorem statements at various points in the development; in some cases these drafts were adopted with revision, in others they were rejected or replaced.

Exposition. Claude was used to assist with drafting, restructuring, and checking revised passages of mathematical exposition. No AI-generated text was incorporated without review and revision by the author.

Responsibility. The overall research programme, the framing of results, all decisions about what to accept, defer, or reject, and final responsibility for the correctness of the paper rest with the author. The author does not regard AI assistance as diminishing the paper's validity, but believes readers are entitled to a complete account of how it was produced.

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Lean 4 Verification Status

During peer review, a reviewer produced an independent Lean 4 formalisation of selected results from this paper, including the exponential denominator growth theorem and the weighted-mediant identity. The author is grateful for this contribution, which helped identify and correct several proof gaps during the review process.

The author has chosen not to publish the Lean 4 development at this stage. Accordingly, **this paper makes no claim of Lean 4 certification or machine-checked verification**. Every theorem, lemma, and proof in this paper stands solely on its hand-argued proof as written in the text. Readers should not infer formal verification status from any part of this manuscript. A complete Lean 4 formalisation of the results in this paper is left for future work.

Reviewer's Comments

The following items identify open proof obligations, gaps, and vulnerabilities in this manuscript. Each item records its source, the manuscript version in which it was raised, and its current status. **Raised** items are newly identified and unaddressed. **Pending** items are scheduled for the next revision. **Deferred** items have been noted but action intentionally postponed. Reviewers should not expect Deferred items to be actioned in the next revision; raising new comments solely about the content of Deferred items is unproductive. **Resolved** items are addressed and awaiting reviewer confirmation. **Rejected** items have had their proposed resolution rejected and must return to **Pending** or **Raised** before the next review. **Dismissed**

items are those for which the author has determined no further action is warranted. **Closed** items are fully accepted by all parties. Only **Dismissed** and **Closed** items are candidates for removal from future versions.

ID	Title	Status
RC-1	Thm. 10.7 ($\Rightarrow$): nested induction unstated	Resolved
RC-2	Thm. 10.7 ($\Leftarrow$): $x < y$ to crossing gap	Closed
RC-3	Thm. 10.5: $d = 0$ base case incomplete	Resolved
RC-4	Thm. 12.6: "Agreement" step unproved	Resolved
RC-5	Lean 4 scope boundary ambiguous	Closed
RC-6	$O(\log N)$ conflates iteration count with bit-op cost	Closed
RC-13	Weight propagation bound slackness uncharacterised	Deferred
RC-14	Base-specific structural dependencies limit generality	Deferred
RC-15	High notation density may impede referee comprehension	Deferred
RC-16	Inter-paper circular coupling: Paper 76 errors propagate to Paper 77	Deferred
RC-21	Conj. 10.9 promoted to Theorem 10.9 with hand proof	Resolved
RC-22	Conj. 10.10 (bound via e) refuted; replaced by Conj. 10.10 and 10.11	Deferred
N1	"Proved in §10" cited but no proof present; now Theorem 10.9	Resolved
N2	Figure 2 caption: misleading description of dashed line and curve ordering	Resolved
N3	Thm. 10.5 $d = 0$: "vacuously" is wrong (= RC-3)	Raised
N4	Thm. 10.7 ($\Rightarrow$): wrong direction cited (= RC-1)	Raised
N5	Thm. 12.6 Agreement: boundary $m/n = o/L$ not excluded (= RC-4)	Raised
N13	Thm. 5.2 stated for $w \geq 1$ but used with $w^- = 0$; extended to $w \geq 0$	Resolved
N14	RC-13–RC-16 refer to content not in the manuscript; noted and deferred	Resolved
N15	Def. 9.2 / Lemma 9.3 / Thm. 10.7: four notation slips	Resolved
N16	§1 "$O(\log N)$ iterations" missing "real-arithmetic" qualifier	Resolved
N17	Ticket uses "Thm 10.8" and "Figure 10.1" but PDF has Thm 10.7 / Fig. 2	Deferred
N18	Def. 6.3: mediant defined as 3-tuple "treated as a ratio"	Resolved
N19	`lakefile.toml`: Round12/13 not default targets; Round8 double-listed	Resolved

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RC-1. Theorem 10.7 ($\Rightarrow$): nested induction unstated. *Source: claude — Since: v0.14 — Status: Resolved*

The $w^-_{d-1} \geq 1$ bullet now cites Theorem 10.3, but the $w^-_{d-1} = 0$ bullet is still unrepaired. It argues: "if r_{2d-2} were a semi-convergent, then by ($\Leftarrow$) at depth $d-1$ it would be the crossing child". That invokes the *converse* of ($\Leftarrow$). The needed direction is "semi-convergent $\Rightarrow$ crossing child", which is the contrapositive of ($\Rightarrow$) at depth $d-1$, i.e. the induction hypothesis. Fix: structure the proof as explicit induction on d, cite ($\Rightarrow$) at depth $d-1$, then use $o_{2d-1} < o_{2d-2}$ and r_{2d-1} better to conclude $w^+_{d-1} \geq 2$.

RC-2. Theorem 10.7 ($\Leftarrow$): inference from $x < y$ to crossing. *Source: claude — Since: v0.14 — Status: Closed*

The ($\Leftarrow$) direction now explicitly invokes the bounded-pair bracketing property (Theorem 7.1) to establish that the signed errors of r^- and r^+ have opposite signs, then computes

the signed error of the unweighted mediant $r^- \oplus r^+$ as $y - x > 0$, making the continuation of the SB path toward r^+ a direct consequence rather than an assertion.

RC-3. Theorem 10.5: $d = 0$ base case incomplete. *Source: claude — Since: v0.14 — Status: Resolved*

The ratio computation giving $w_0^+ \geq 2$ is now complete. However, the conclusion "the statement holds vacuously at $d = 0$" is wrong: with $w_0^+ \neq 1$ the biconditional requires r_1 *not* to be a semi-convergent, and that requires an argument (Remark 10.8: if $\lceil\theta\rceil$ is the better leaf then $\{\theta\} > \frac{1}{2}$, so $a_1 = 1$ and $\lceil\theta\rceil = p_1/q_1$ is the first convergent). Fix: move or inline Remark 10.8 before Theorem 10.5, or supply the argument directly. Also, $|A| = |\varepsilon_{r_0}|$ conflates absolute error with relative error; they differ by θ, which is harmless for the ratio but should be noted.

RC-4. Theorem 12.6: "Agreement" step unproved. *Source: claude — Since: v0.14 — Status: Resolved*

The Key lemma is now invoked, but the boundary case is not excluded. The integer m with $n\alpha < m \leq n \cdot o/L$ may satisfy $m = n \cdot o/L$, in which case $m/n = o/L$ is *not* strictly between α and o/L. Fix: add "$m/n = o/L$ is impossible since $\gcd(o, L) = 1$ and $0 < n < L$; and $m = n\alpha$ is impossible since α is irrational".

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RC-5. Lean 4 scope boundary ambiguous. *Source: gemini — Since: v0.14 — Status: Resolved*

All Lean 4 citations have been removed from the paper. The paper makes no claim of machine-checked or formally verified status. Every result stands solely on the hand-argued proofs in the text. A full Lean development is not published at this stage and is left for future work.

RC-6. $O(\log N)$ iteration count conflates steps with oracle cost. *Source: gemini — Since: v0.14 — Status: Resolved*

The paper now claims only that $O(\log N)$ real-arithmetic iterations suffice to reach an approximant of size N, which is what the convergence theorem (Section 10) and Conjecture 10.9 assert. No claim is made about bit-operation cost per oracle call; that is a separate question outside the scope of this paper.

RC-13. Weight propagation bound slackness uncharacterised. *Source: gemini — Since: v0.14 — Status: Deferred*

Note (N14): the phrase "trajectory spikes" does not correspond to any concept in the manuscript. The paper makes no asymptotic claims that depend on weight-propagation slackness; the relevant complexity bound is now Theorem 10.9.

RC-14. Base-specific structural dependencies limit generality. *Source: gemini — Since: v0.14 — Status: Deferred*

Note (N14): the paper does not employ 2-adic or 3-adic valuations, nor does it treat generalised Collatz variants. These concepts are not present in the manuscript; the comment appears to be directed at a different paper.

RC-15. High notation density may impede referee comprehension. *Source: gemini — Since: v0.14 — Status: Deferred*

Note (N14): the phrase "stream-tagging frameworks" does not appear in the manuscript. The density concern may be legitimate in principle but the specific evidence cited is not drawn from this paper.

RC-16. Inter-paper circular coupling: Paper 76 errors propagate to Paper 77. *Source: gemini — Since: v0.14 — Status: Deferred*

Note (N14): this comment rests on the hallucinated content of RC-13–RC-15. The cross-paper dependency between Papers 76 and 77 is real and acknowledged, but no "circular coupling" exists: Paper 77 uses definitions from Paper 76 unidirectionally, and the errors identified in the review process were independent in both papers.

RC-21. Conjecture 10.9 proved; universal constants. *Source: author — Since: v0.14 — Status: Resolved*

Conjecture 10.9 has been promoted to Theorem 10.9 with a hand proof (Lemma A: no two consecutive unit weights; Lemma B: every denominator at depth d is at least $2^{\lfloor d/2 \rfloor}$; conclusion: $d \leq 2\log_2(e_d + o_d) + 1$). The constants $m = 2$, $b = 1$ are universal and independent of θ. All forward references in the Abstract, §1, and §12 now cite Theorem 10.9 directly.

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RC-22. Original Conjecture 10.10 (universal bound via e) refuted; replaced. *Source: gemini — Since: v0.14 — Status: Deferred*

The original conjecture that the constants (m_e, b_e) for $\theta = e$ serve as universal bounds is false and ill-posed. Aristotle (`golden_violates_small_slope`) proves that for $\theta = \phi$ no line $m \cdot \log_2(e_d + o_d) + b$ with $m \leq 0.7$ bounds the depth for all d, regardless of b. The crossover is not at extreme depth: at $d = 5$ the pair for ϕ is $233/144$ and $0.43 \cdot \log_2 377 + 1.19 < 5$ already (`golden_depth5_crossover`). Furthermore e is asymptotically *near the bottom* of the family, not the top, because its weights grow without bound. The original conjecture has been replaced by Conjectures 10.10 and 10.11, which separate the existence of a sub-universal bound from the identification of ϕ as the worst case. Pending: resolve both new conjectures.

N1. "Proved in §10" cited but no such proof present. *Source: aristotle — Since: v0.14 — Status: Resolved*

Resolved by RC-21. Conjecture 10.9 is now Theorem 10.9 with a complete hand proof in §10. All forward references in the Abstract, §1, and §12 now point to Theorem 10.9 directly.

N2. Figure caption: misleading description of the dashed line and curve ordering. *Source: aristotle — Since: v0.14 — Status: Resolved*

The caption contained three errors that could mislead a reader into thinking the dashed line was a universal bound and that ϕ lay below e: (a) it said ϕ "sits between e and the logarithmic targets", whereas ϕ and $\sqrt{2}$ both lie *above* e in the plotted range; (b) it said "Slope varies: π and $\log_3 8$ grow fastest" — the opposite of what the figure shows: π and $\log_3 8$ have the *shallowest* slopes (fastest denominator growth due to large partial quotients); (c) by saying e is consistent with the (then-stated) universal bound conjecture immediately after describing the dashed line, it implied the dashed line was meant to bound all curves. The caption has been rewritten to state explicitly that the dashed line bounds $\log_2 3$ only, and to correctly describe the ordering of the curves.

N13. Theorem 5.2 stated for $w \geq 1$ but used with $w^- = 0$. *Source: aristotle — Since: v0.14 — Status: Resolved*

Theorem 5.2 (mediant convergence) has been extended to $w \geq 0$. The $w = 0$ case ($r^- = r_{2d}$, the parent) is trivial and noted explicitly in the statement; the error bounds hold with equality on the left.

N14. RC-13–RC-16 refer to content not present in the manuscript. *Source: aristotle — Since: v0.14 — Status: Resolved*

RC-13–RC-16 remain Deferred; each cites concepts not present in the manuscript. Each cites concepts not present in this manuscript: "trajectory spikes" (RC-13), "2-adic and 3-adic valuations" and generalised Collatz variants (RC-14), "stream-tagging frameworks" (RC-15). RC-16's "circular coupling" claim rests on these hallucinated items. The comments appear to have been generated against a different paper or a conflated description of multiple papers.

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N15. Definition 9.2 / Lemma 9.3 / Theorem 10.7: notation slips. *Source: aristotle — Since: v0.14 — Status: Resolved*

All four slips fixed: (a) Definition 9.2 now says "propagated optimal weight of the parent pair"; (b) the closing paragraph of Lemma 9.3 now explicitly states $u = \pi(r_{2d})$, $v = \pi(r_{2d+1})$ at depth d, consistent with the $w_d^{\pm}$ indexing; (c) Definition 9.2 now notes that Δ is the determinant of the parent pair; (d) Theorem 10.7 no longer uses r_{d-1}^+; replaced by the explicit expression $r_{2d-2} + w_{d-1}^+ r_{2d-1} = r_{2d+1}$.

N16. §1 missing "real-arithmetic" qualifier on $O(\log N)$. *Source: aristotle — Since: v0.14 — Status: Resolved*

Resolved. Section 1 item 4 now reads "$O(\log N)$ real-arithmetic iterations", consistent with the qualifier applied throughout the rest of the paper in response to RC-6.

N17. Ticket numbering mismatches the PDF. *Source: aristotle — Since: v0.14 — Status: Deferred*

The Round 14 review ticket refers to "Thm 10.8" (the PDF has Thm 10.7; 10.8 is a Remark), "Figure 10.1" (the PDF has Figure 2), and quotes the bounding-line parameters as 0.454/1.92 while the PDF prints 0.45/1.9. This is an organisational issue with how future

review tickets are generated, not a defect in the paper itself. Deferred to the review-process workflow.

N18. Definition 6.3: mediant defined as a 3-tuple "treated as a ratio". *Source: aristotle — Since: v0.14 — Status: Resolved*

Resolved. Definition 6.3 now defines $\mathrm{Mediant}(r, r', w) = (e + we', o + wo')$ directly as a ratio, dropping the awkward "3-tuple treated as a ratio" language.

N19. `lakefile.toml`: inconsistent default targets. *Source: aristotle — Since: v0.14 — Status: Resolved*

Resolved. The proposed `lakefile.toml` (delivered in Round 14 as `lakefile.proposed.toml`) has been applied: each library now owns exactly one module; `Round12_MT`, `Round12_76`, and `Round13_76` are now listed as default targets; the duplicate `Round8_76`/`Round8_77` glob entries are removed.

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